

# Vector Calculus

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## Lecture 1 – 23/01/26

### §1 Preliminaries

#### §1.1 Vector notation

We will in general work in  $\mathbb{R}^3$ , but sometimes  $\mathbb{R}^2$ . In fact, most of our results will generalise to  $\mathbb{R}^n$ .

We will write column vectors to save space when we mean row vectors, and we will use dot notation for differentiation (even though  $t$  isn't really time).

If  $\mathbf{a}$  and  $\mathbf{b}$  are vectors, then the following hold:

$$\dot{\mathbf{a}}(t) = \frac{d\mathbf{a}}{dt} = \frac{d}{dt}(a_1, a_2, a_3) = (\dot{a}_1, \dot{a}_2, \dot{a}_3) \quad (1)$$

$$\frac{d}{dt}(\mathbf{a} \cdot \mathbf{b}) = \dot{\mathbf{a}} \cdot \mathbf{b} + \mathbf{a} \cdot \dot{\mathbf{b}} \quad (2)$$

$$\frac{d}{dt}(\mathbf{a} \times \mathbf{b}) = \dot{\mathbf{a}} \times \mathbf{b} + \mathbf{a} \times \dot{\mathbf{b}} \quad (3)$$

and these results can be shown using suffix notation. Integration is also componentwise. We have therefore

$$\int_{t_1}^{t_2} \dot{\mathbf{a}}(t) dt = \mathbf{a}(t_2) - \mathbf{a}(t_1)$$

The modulus of a vector will be denoted by the same letter as the vector, but not bolded/underlined. So, for example,  $r = |\mathbf{r}|$  (but  $x$  means  $x_1$ , not  $|\mathbf{x}|$ ).

Notationally, we might write  $\partial_i$  for  $\frac{\partial}{\partial x_i}$ .

#### §1.2 The chain rule

Recall that, if  $f$  is a function of  $\mathbf{x}$  which is itself a function of  $t$ , then

##### Proposition 1.1

$$\begin{aligned} \frac{d}{dt}f(\mathbf{x}(t)) &= \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt} \\ &= \frac{\partial f}{\partial x_i} \frac{dx_i}{dt} \end{aligned}$$

If instead  $\mathbf{x} = \mathbf{x}(\mathbf{u})$ , where  $\mathbf{u} = (u_1, u_2, u_3) = (u, v, w)$ , then

$$\frac{\partial f}{\partial u} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial u} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial u}$$

and similarly for  $\frac{\partial f}{\partial v}$  and  $\frac{\partial f}{\partial w}$ , i.e.

$$\frac{\partial f}{\partial u_i} = \frac{\partial f}{\partial x_i} \frac{\partial x_i}{\partial u_i}$$

### §1.3 Taylor's theorem

Recall that for a small vector displacement  $\delta \mathbf{x}$ ,

$$\begin{aligned} f(\mathbf{x} + \delta \mathbf{x}) &= f(x + \delta x, y + \delta y, z + \delta z) \\ &= f(x, y, z) + \frac{\partial f}{\partial x} \delta x + \frac{\partial f}{\partial y} \delta y + \frac{\partial f}{\partial z} \delta z + \frac{1}{2!} \left( \delta x \frac{\partial}{\partial x} + \delta y \frac{\partial}{\partial y} + \delta z \frac{\partial}{\partial z} \right)^2 f + \dots \end{aligned}$$

or, in suffix notation,

#### Proposition 1.2

$$f(\mathbf{x} + \delta \mathbf{x}) = f(\mathbf{x}) + \frac{\partial f}{\partial x_i} \delta x_i + O(|\delta \mathbf{x}|^2)$$

### §1.4 Infinitesimals

In one dimension, we have from Taylor's theorem

$$\delta f = f(x + \delta x) - f(x) = f'(x) \delta x + O(\delta x^2)$$

Taking  $\delta x \rightarrow 0$ ,  $\delta x$  and  $\delta f$  become  $dx$  and  $df$ , and we get the exact equality

$$df = f'(x) dx$$

If we want, we can even think of infinitesimals as numbers such that their squares are zero.

Note that infinitesimals have their own product and quotient rules, namely

$$\begin{aligned} d(fg) &= g df + f dg \\ d\left(\frac{f}{g}\right) &= \frac{g df - f dg}{g^2} \end{aligned}$$

We also get

#### Proposition 1.3

$$df = \frac{\partial f}{\partial x_i} dx_i$$

The modulus of an infinitesimal vector can be calculated using, for example

$$|d\mathbf{x}| = |dx| \sqrt{1 + \left(\frac{dy}{dx}\right)^2 + \left(\frac{dz}{dx}\right)^2}$$

If  $dx$  and  $dy$  are independent infinitesimals then  $dx dy$  is an infinitesimal area. But if  $y$  depends on  $x$  in some way then  $dx dy$  vanishes, since it's proportional to  $dx^2$ .

## §2 Vector differential operators

**Remark.** What is an operator? It is a map between vector spaces of functions.

## §2.1 Del

The fundamental differential operator is *del*:

$$\nabla = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix}$$

We can also write

$$\nabla = \mathbf{e}_1 \frac{\partial}{\partial x_1} + \mathbf{e}_2 \frac{\partial}{\partial x_2} + \mathbf{e}_3 \frac{\partial}{\partial x_3} = \mathbf{e}_i \frac{\partial}{\partial x_i}$$

where  $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$  is the canonical basis for  $\mathbb{R}^3$ .

## §2.2 Gradient

When del is applied to a scalar field  $f(\mathbf{x})$ , it is known as the *gradient* operator:

$$\nabla f = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \\ \frac{\partial f}{\partial z} \end{pmatrix}$$

This can also be denoted  $\text{grad} f$ .

In suffix notation, we have

$$\begin{aligned} (\nabla f)_i &= \frac{\partial f}{\partial x_i} \\ \iff \nabla f &= \frac{\partial f}{\partial x_i} \mathbf{e}_i \end{aligned}$$

**Example** • If  $f(\mathbf{x}) = |\mathbf{x}|^2 = x^2 + y^2 + z^2$ , then  $\nabla f = (2x, 2y, 2z) = 2\mathbf{x}$ . Alternatively,  $\frac{\partial}{\partial x_i}(x_j x_j) = 2x_i \delta_{ij} = 2x_i$  (the  $\delta_{ij}$  comes from  $\frac{\partial x_j}{\partial x_i}$ ).

- $\nabla |\mathbf{x} - \mathbf{x}_0|^2 = 2(\mathbf{x} - \mathbf{x}_0)$ . This is a direct consequence of the above and a very simple application of the chain rule.
- $\nabla(\mathbf{a} \cdot \mathbf{x}) = \nabla(a_1 x + a_2 y + a_3 z) = a_1 \mathbf{e}_1 + a_2 \mathbf{e}_2 + a_3 \mathbf{e}_3 = \mathbf{a}$ . Alternatively,  $\frac{\partial}{\partial x_i}(a_j x_j) = a_i \delta_{ij} = a_i$ .

$\nabla f$  in 3D is analogous to  $f'$  in 1D, and we can evaluate it at points.

**Example**

If  $f(\mathbf{x}) = x + y^2 + z^4$ , then  $\nabla f(\mathbf{a}) = \nabla f(\mathbf{b})$  if and only if  $\mathbf{a} - \mathbf{b}$  is parallel to the  $x$ -axis. Observe  $\nabla f(\mathbf{x}) = (1, 2y, 4x^3)$ , so

$$\begin{aligned} (1, 2a_2, 4a_3^3) &= (1, 2b_2, 4b_3^3) \\ \iff (a_2, a_3) &= (b_2, b_3) \\ \iff \mathbf{a} - \mathbf{b} &= (a_1 - b_1, 0, 0) \end{aligned}$$

We have some new opportunities for notation now – for example, we can rewrite (1.1) as

**Proposition 2.1**

$$\frac{d}{dt}f(\mathbf{x}(t)) = \dot{\mathbf{x}} \cdot \nabla f$$

and (1.3) as

**Proposition 2.2**

$$df = \nabla f \cdot d\mathbf{x}$$

Equation (2.2) also gives a way of defining  $\nabla f$  that is *coordinate-independent*, i.e. it doesn't require us to choose a particular basis at any point.

**Example**

If  $f(x) = |\mathbf{x}|^2$ , then

$$\begin{aligned} df &= f(\mathbf{x} + d\mathbf{x}) - f(\mathbf{x}) = \mathbf{x} \cdot \mathbf{x} + 2\mathbf{x} \cdot d\mathbf{x} + d\mathbf{x} \cdot d\mathbf{x} - |\mathbf{x}|^2 \\ &= 2\mathbf{x} \cdot d\mathbf{x} \end{aligned}$$

so it must be the case that  $\nabla f = 2\mathbf{x}$ .

Finally, we may write  $\nabla_{\mathbf{x}}$ ,  $\nabla_{\mathbf{y}}$ , etc., to denote differentiation w.r.t. a particular vector variable.

**§2.3 Divergence**

**Definition.** The *divergence* of a vector field is given by del dotted with the field:

$$\nabla \cdot \mathbf{F} = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \cdot \begin{pmatrix} F_1 \\ F_2 \\ F_3 \end{pmatrix} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} = \frac{\partial F_i}{\partial x_i}$$

More formally, we can derive this expression by

$$\nabla \cdot \mathbf{F} = \left( \mathbf{e}_i \frac{\partial}{\partial x_i} \right) \cdot (F_j \mathbf{e}_j)$$

$$\begin{aligned}
&= \mathbf{e}_i \cdot \frac{\partial}{\partial x_i} (F_j \mathbf{e}_j) \\
&= \mathbf{e}_i \cdot \frac{\partial F_j}{\partial x_i} \mathbf{e}_j \\
&= \frac{\partial F_j}{\partial x_i} \delta_{ij} = \frac{\partial F_i}{\partial x_i}
\end{aligned}$$

What is the divergence? It measures the amount by which a field is ‘expanding’ around  $\mathbf{x}$ . We will get a better idea later on when we derive a coordinate-independent definition.

**Example** • Take  $F(\mathbf{x}) = \mathbf{x}$ . Then  $\nabla \cdot F = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 3$ . Suffix notation also works! DIAGRAM

- Take  $F(\mathbf{x}) = (-y, x, 0)$ . Then  $\nabla \cdot F = -\frac{\partial y}{\partial x} + \frac{\partial x}{\partial y} = 0$ . DIAGRAM
- If  $A$  is a 3 by 3 matrix, we might want to calculate

$$\nabla \cdot (A\mathbf{x}) = \frac{\partial x}{\partial x_i} (A_{ij} x_j) = A_{ij} \delta_{ij} = A_{ii} = \text{Tr } A$$

## §2.4 Curl

**Definition.** The *curl* of a vector field is given by del crossed with the field:

$$\nabla \times \mathbf{F} = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \times \begin{pmatrix} F_1 \\ F_2 \\ F_3 \end{pmatrix} = \begin{pmatrix} \frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \\ \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \\ \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \end{pmatrix}$$

In suffix notation, we have

$$(\nabla \times F)_i = \varepsilon_{ijk} \frac{\partial}{\partial x_j} F_k$$

The curl, notably, is only defined in 3D, since we only have a cross product in 3D, unlike gradient or divergence. It measures, in a certain sense, the *rotation* of the field about a point  $\mathbf{x}$ .

**Example** • With  $F(\mathbf{x}) = \mathbf{x}$ , we have

$$\nabla \times F = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \times \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \mathbf{0}$$

Alternatively,  $(\nabla \times F)_i = \varepsilon_{ijk} \frac{\partial}{\partial x_j} x_k = \varepsilon_{ijk} \delta_{jk} = 0$ .

- With  $F(\mathbf{x}) = (-y, x, 0)$ ,  $\nabla \times F = (0, 0, 2)$ .
- If  $A$  is a 3 by 3 symmetric matrix,

$$(\nabla \times (A\mathbf{x}))_i = \varepsilon_{ijk} \frac{\partial}{\partial x_j} A_{kl} x_l = \varepsilon_{ijk} A_{kj}$$

But  $\varepsilon_{ijk}$  is antisymmetric while  $A_{kj}$  is symmetric, so the terms cancel in pairs and we get  $\nabla \times (A\mathbf{x}) = \mathbf{0}$ .

## §2.5 Directional derivatives

Given a *vector* field  $\mathbf{u}$  we can define a scalar *operator*

$$\mathbf{u} \cdot \nabla = u_1 \frac{\partial}{\partial x_1} + u_2 \frac{\partial}{\partial x_2} + u_3 \frac{\partial}{\partial x_3} = u_i \frac{\partial}{\partial x_i}$$

We can then apply this operator to a *scalar* field  $f$ :

$$(\mathbf{u} \cdot \nabla)f = u_1 \frac{\partial f}{\partial x_1} + u_2 \frac{\partial f}{\partial x_2} + u_3 \frac{\partial f}{\partial x_3} = u_i \frac{\partial f}{\partial x_i}$$

But, happily, it turns out

$$(\mathbf{u} \cdot \nabla)f = \left( u_i \frac{\partial}{\partial x_i} \right) f = u_i \frac{\partial f}{\partial x_i} = \mathbf{u} \cdot (\nabla f)$$

and so we can omit the brackets.

However, we can also apply  $\mathbf{u} \cdot \nabla$  to a *vector* field  $\mathbf{F}$ :

$$(\mathbf{u} \cdot \nabla)\mathbf{F} = u_1 \frac{\partial}{\partial x_1} \mathbf{F} + u_2 \frac{\partial}{\partial x_2} \mathbf{F} + u_3 \frac{\partial}{\partial x_3} \mathbf{F}$$

so that

$$((\mathbf{u} \cdot \nabla)\mathbf{F})_i = u_j \frac{\partial F_i}{\partial x_j}$$

and this gives us a *vector* field back. Disturbingly, the brackets here *are* important, since  $\mathbf{u} \cdot (\nabla \mathbf{F})$  is the gradient of a vector which isn't defined. However, we omit them anyway (since there's no ambiguity).

**Example**

Take  $\mathbf{u} = (0, x, y)$  and  $\mathbf{F} = (yz, 0, x)$ . Then  $\mathbf{u} \cdot \nabla = x \frac{\partial}{\partial y} + y \frac{\partial}{\partial z}$ , so  $(\mathbf{u} \cdot \nabla)\mathbf{F} = (xz + y^2, 0, 0)$ .

**Definition.** Given a unit vector  $\hat{\mathbf{n}}$ , the *directional derivative*  $D_{\hat{\mathbf{n}}}f$  of  $f$  at  $\mathbf{x}$  is defined to be the one-dimensional derivative of  $f$  along the line from  $\mathbf{x}$  parallel to  $\hat{\mathbf{n}}$ . That is, if we make  $g(t) = f(\mathbf{x} + t\hat{\mathbf{n}})$ , then

$$D_{\hat{\mathbf{n}}}f(\mathbf{x}) = g'(0) = \hat{\mathbf{n}} \cdot \nabla f(\mathbf{x})$$

by [proposition 2.1](#). Then  $D_{\hat{\mathbf{n}}} = \hat{\mathbf{n}} \cdot \nabla$ , and so the operator  $\mathbf{u} \cdot \nabla$  is also called the directional derivative (even for non-unit vectors  $\mathbf{u}$ ).

**§2.6 Level surfaces and the gradient**

**Definition.** In 3D, a *level surface* of a function  $f(\mathbf{x})$  is a two-dimensional surface satisfying  $f(\mathbf{x}) = c$  for some constant  $c$ . In 2D, these are just curves which we refer to as *contours*.

**Proposition**

The gradient of  $f$  is always normal to a level surface.

*Proof.* Starting at  $\mathbf{x}$  on the surface, consider a displacement  $d\mathbf{x}$  within the surface. Because  $f = c$ ,  $df = 0$  for this displacement, so, by [proposition 2.2](#),  $\nabla f \cdot d\mathbf{x} = df = 0$  for all  $d\mathbf{x}$  lying in the surface. So  $\nabla f$  is perpendicular to all of these and hence is perpendicular to the surface. ■

**Example**

Take  $f(\mathbf{x}) = |\mathbf{x}|^2$ , so  $\nabla f = 2\mathbf{x}$ . Since this is simply radially outwards, we conclude the level surfaces are spheres centred at the origin.

**Proposition**

$\nabla f$  always points in the direction in which  $f$  grows the most rapidly.

*Proof.* Consider  $D_{\hat{\mathbf{n}}}f = \hat{\mathbf{n}} \cdot \nabla f$  for an arbitrary unit vector  $\hat{\mathbf{n}}$ . Clearly  $\hat{\mathbf{n}} \cdot \nabla f$  is maximised when  $\hat{\mathbf{n}}$  and  $\nabla f$  are parallel, and in particular this maximum is  $|\nabla f|$ . ■

**Remark.** We can think of level surfaces in two different ways, and endow them with different grads. Suppose we have a ground level  $z = h(x, y)$  (imagine hills). Working in 2D, we have that  $\nabla h$  points in the direction that is steepest uphill (or the projection of that onto the  $x$ - $y$  plane). On the other hand, if we consider  $f(x, y, z) = z - h(x, y)$  (which defines the exact same surface), then the level surface  $f = 0$  has  $\nabla f$  pointing normal to the surface.

## §2.7 Various vector differential identities

The following important identities can be proven using suffix notation.

### Proposition 2.3

$$\nabla(fg) = f\nabla g + g\nabla f$$

### Proposition 2.4

$$\nabla(f(g(\mathbf{x}))) = f'(g(\mathbf{x}))\nabla g$$

### Proposition 2.5

$$\nabla \cdot (f\mathbf{F}) = f\nabla \cdot \mathbf{F} + \mathbf{F} \cdot \nabla f$$

*Proof.* We have

$$\begin{aligned}\nabla \cdot (f\mathbf{F}) &= \frac{\partial}{\partial x_i}(fF_i) \\ &= \frac{\partial f}{\partial x_i}F_i + f\frac{\partial F_i}{\partial x_i} \\ &= \nabla f \cdot \mathbf{F} + f\nabla \cdot \mathbf{F}\end{aligned}$$

■

### Proposition 2.6

$$\nabla \times (f\mathbf{F}) = f\nabla \times \mathbf{F} + \nabla f \times \mathbf{F}$$

The above propositions should be memorised – it is not necessary to memorise the following propositions, but one should be able to produce proofs of them.

### Proposition 2.7

$$\nabla(\mathbf{F} \cdot \mathbf{G}) = \mathbf{F} \times (\nabla \times \mathbf{G}) + (\mathbf{F} \cdot \nabla)\mathbf{G} + \mathbf{G} \times (\nabla \times \mathbf{F}) + (\mathbf{G} \cdot \nabla)\mathbf{F}$$

*Proof.* Counterintuitively, let us start with

$$\begin{aligned}(\mathbf{F} \times (\nabla \times \mathbf{G}))_i &= \varepsilon_{ijk}F_j\varepsilon_{klm}\frac{\partial}{\partial x_l}G_m \\ &= (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl})F_j\frac{\partial G_m}{\partial x_l}\end{aligned}$$

$$= F_j \frac{\partial G_j}{\partial x_i} - F_j \frac{\partial G_i}{\partial x_j}$$

Adding to a similar expression for  $(\mathbf{G} \times (\nabla \times \mathbf{F}))_i$ , we get

$$\begin{aligned} & F_j \frac{\partial G_j}{\partial x_i} - F_j \frac{\partial G_i}{\partial x_j} + G_j \frac{\partial F_j}{\partial x_i} - G_j \frac{\partial F_i}{\partial x_j} \\ &= \underbrace{F_j \frac{\partial G_j}{\partial x_i} + G_j \frac{\partial F_j}{\partial x_i}}_{\frac{\partial}{\partial x_i}(F_j G_j)} - F_j \frac{\partial G_i}{\partial x_j} - G_j \frac{\partial F_i}{\partial x_j} \\ &= (\nabla(\mathbf{F} \cdot \mathbf{G}))_i - ((\mathbf{F} \cdot \nabla)\mathbf{G})_i - ((\mathbf{G} \cdot \nabla)\mathbf{F})_i \end{aligned}$$

and the result follows. ■

### Proposition 2.8

$$\nabla \cdot (\mathbf{F} \times \mathbf{G}) = -\mathbf{F} \cdot (\nabla \times \mathbf{G}) + \mathbf{G} \cdot (\nabla \times \mathbf{F})$$

### Proposition 2.9

$$\nabla \times (\mathbf{F} \times \mathbf{G}) = (\nabla \cdot \mathbf{G})\mathbf{F} - (\nabla \cdot \mathbf{F})\mathbf{G} + (\mathbf{G} \cdot \nabla)\mathbf{F} - (\mathbf{F} \cdot \nabla)\mathbf{G}$$

**Remark.** We must not use the formulae  $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}$  or  $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$ , since our derivations assumed certain commutativity and anti-commutativity properties that don't hold for operators.

We have one more very useful identity.

### Proposition 2.10

$$\frac{\partial r}{\partial x_i} = \frac{x_i}{r}$$

*Proof.* We have  $r^2 = x_j x_j$ , so  $2r \frac{\partial r}{\partial x_i} = 2x_j \frac{\partial x_j}{\partial x_i} = 2x_j \delta_{ij} = 2x_i$ . ■

### Corollary 2.11

$$\nabla f(r) = f'(r)\mathbf{e}_r$$

where  $\mathbf{e}_r = \mathbf{r}/r$ .

*Proof.* We have

$$\frac{\partial}{\partial x_i} f(r) = f'(r) \frac{\partial r}{\partial x_i} = f'(r) \frac{x_i}{r}$$



### Example

Using [proposition 2.5](#) and [corollary 2.11](#), we have

$$\begin{aligned}
 \nabla \cdot (r^3 \mathbf{r}) &= r^3 \nabla \cdot \mathbf{R} + \mathbf{r} \cdot \nabla r^3 \\
 &= 3r^3 + \mathbf{r} \cdot 3r^2 \mathbf{e}_r \\
 &= 3r^3 + \mathbf{r} \cdot 3r\mathbf{r} \\
 &= 6r^3
 \end{aligned}$$

To deal with the modulus of a vector field, we can do

$$\frac{\partial}{\partial x_i} |\mathbf{F}| = \frac{\partial}{\partial x_i} \sqrt{F_j F_j} = \frac{1}{2|\mathbf{F}|} \frac{\partial}{\partial x_i} (F_j F_j) = \frac{1}{|\mathbf{F}|} F_j \frac{\partial F_j}{\partial x_i}$$

## §2.8 The Laplacian

**Definition.** The *Laplacian* of a scalar field is

$$\begin{aligned}
 \nabla^2 f &= \nabla \cdot (\nabla f) \\
 &= \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \\
 &= \frac{\partial^2 f}{\partial x_i \partial x_i}
 \end{aligned}$$

It may also be denoted  $\Delta f$ .

**Example** • If  $f(\mathbf{x}) = \alpha x^2 + \beta y^2 + \gamma z^2$ , then

$$\left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) (\alpha x^2 + \beta y^2 + \gamma z^2) = 2(\alpha + \beta + \gamma)$$

- $\nabla^2 |\mathbf{x}|^2 = \nabla^2 (x^2 + y^2 + z^2) = 6$ .
- If  $f(\mathbf{x}) = (\mathbf{a} \cdot \mathbf{x})^2$ ,

$$\begin{aligned} \nabla^2 f &= \nabla \cdot (\nabla f) \\ &= \nabla \cdot (2(\mathbf{a} \cdot \mathbf{x}) \nabla (\mathbf{a} \cdot \mathbf{x})) \\ &= \nabla \cdot (2(\mathbf{a} \cdot \mathbf{x}) \mathbf{a}) \\ &= 2(\mathbf{a} \cdot \mathbf{x}) \nabla \cdot \mathbf{a} + 2\mathbf{a} \cdot \nabla (\mathbf{a} \cdot \mathbf{x}) \\ &= 2\mathbf{a} \cdot \mathbf{a} \\ &= 2|\mathbf{a}|^2 \end{aligned}$$

Alternatively, and perhaps more easily,

$$\begin{aligned} \nabla^2 f &= \nabla^2 (a_1^2 x^2 + a_2^2 y^2 + a_3^2 z^2 + 2a_1 a_2 xy + 2a_1 a_3 xz + 2a_2 a_3 yz) \\ &= 2(a_1^2 + a_2^2 + a_3^2) \\ &= 2|\mathbf{a}|^2 \end{aligned}$$

Functions that satisfy  $\nabla^2 f = 0$  (Laplace's equation) are called *harmonic*.

We often have spherically symmetric functions, i.e. those for which  $f = f(r)$ . In that case,

$$\begin{aligned} \nabla^2 f(r) &= \nabla \cdot (\nabla f(r)) \\ &= \nabla \cdot (f'(r) \mathbf{e}_r) \\ &= \frac{\partial}{\partial x_i} \left( \frac{f'(r)}{r} x_i \right) \\ &= \frac{r f''(r) - f'(r)}{r^2} \frac{x_i}{r} x_i + \frac{f'(r)}{r} \delta_{ii} \\ &= f''(r) + \frac{2f'(r)}{r} \end{aligned}$$

This can also be written  $\frac{1}{r^2} \frac{d}{dr} (r^2 f'(r))$  or  $\frac{1}{r} \frac{d^2}{dr^2} (rf)$ .

**Remark.** As we can see, the only thing that depends on the dimension is  $\delta_{ii}$ . So, if we were in 2D, we would have  $f''(r) + \frac{f'(r)}{r}$  instead.

For example,  $\nabla^2 |\mathbf{x}|^2 = \nabla^2 f$  with  $f(r) = r^2$ , so  $\nabla^2 f = 2 + \frac{4r}{r} = 6$ .

For a vector field  $\mathbf{F}(\mathbf{x})$ , we can't define  $\nabla^2 \mathbf{F}$  as  $\nabla \cdot (\nabla \mathbf{F})$ , since  $\nabla \mathbf{F}$  is undefined. Instead, we say

$$\nabla^2 \mathbf{F} = (\nabla^2 F_1, \nabla^2 F_2, \nabla^2 F_3)$$

An alternative definition that is coordinate-independent is

$$\nabla^2 \mathbf{F} = \nabla(\nabla \cdot \mathbf{F}) - \nabla \times (\nabla \times \mathbf{F})$$

but the first definition is preferable for calculations. To show they are equivalent, we have

$$\begin{aligned} (\nabla \times (\nabla \times \mathbf{F}))_i &= \varepsilon_{ijk} \frac{\partial}{\partial x_j} \varepsilon_{klm} \frac{\partial F_m}{\partial x_l} \\ &= \frac{\partial^2 F_j}{\partial x_i \partial x_j} - \frac{\partial^2 F_i}{\partial x_j \partial x_j} \\ &= \frac{\partial}{\partial x_i} (\nabla \cdot \mathbf{F}) - \nabla^2 (F_i) \end{aligned}$$

## §2.9 Irrotational and conservative vector fields

Consider a vector field  $\mathbf{F}$  defined in a domain  $\mathcal{D} \subseteq \mathbb{R}^3$ .

**Definition.** If  $\mathbf{F}$  satisfies  $\nabla \times \mathbf{F} = \mathbf{0}$  in  $\mathcal{D}$ , it is *irrotational* there.

**Definition.** If instead there exists a single-valued scalar field  $f$  defined in  $\mathcal{D}$  such that  $\mathbf{F} = \nabla f$ , then  $\mathbf{F}$  is said to be *conservative*, and  $f$  is known as a *scalar potential* for  $\mathbf{F}$ . Sometimes the sign of  $f$  is changed, so  $\mathbf{F} = -\nabla f$ .

### Proposition 2.12

A vector field is conservative if and only if it is irrotational.

*Proof.*  $\Rightarrow$  If  $\mathbf{F} = \nabla f$ , then  $\nabla \times \mathbf{F} = \nabla \times (\nabla f)$ . But

$$(\nabla \times (\nabla f))_i = \varepsilon_{ijk} \frac{\partial}{\partial x_i} \frac{\partial f}{\partial x_k}$$

and  $\varepsilon_{ijk}$  is antisymmetric while  $\frac{\partial^2 f}{\partial x_j \partial x_k}$  is symmetric, so the terms cancel in pairs.

$\Leftarrow$  The converse holds only if  $\mathcal{D}$  is *simply connected*. The result will be shown in a later section. ■

**Definition.** A *simply connected domain*  $\mathcal{D}$  is a connected domain in which any closed curve can be shrunk continuously to a point.

**Example** •  $\mathbb{R}^3$  is simply connected.

- $\mathbb{R}^3 \setminus \{(x, 0, 0) : x \in \mathbb{R}\}$  is not simply connected - take a curve wrapped around the  $x$ -axis.
- $\mathbb{R}^3 \setminus \{(x, 0, 0) : x \geq 0\}$  is simply connected.
- $\mathbb{R}^3 \setminus \{\mathbf{0}\}$  is simply connected.

Importantly, consider  $\mathbf{F}(\mathbf{x}) = \frac{1}{x^2+y^2}(-y, x, 0)$ . We can see that  $\mathbf{F}$  is irrotational. We can also see that  $f(\mathbf{x}) = \arctan(y/x)$  satisfies  $\mathbf{F} = \nabla f$  - but  $f$  is not a single-valued function - or, if we choose the principal value, it's not continuous. Indeed, the domain of  $\mathbf{F}$  is not simply connected, since it is undefined on the  $z$ -axis.

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## §2.10 Solenoidal/incompressible vector fields

**Definition.** A vector field  $\mathbf{F}$  is *solenoidal* (or *divergence-free*) in a domain  $\mathcal{D}$  if  $\nabla \cdot \mathbf{F} = 0$  there.

**Remark.** The nomenclature arises from magnetic fields, which are always solenoidal (this is actually equivalent to the non-existence of magnetic monopoles). In fluid dynamics, on the other hand, such fields are called incompressible.

If there exists a vector field  $\mathbf{A}$  in  $\mathcal{D}$  such that  $\mathbf{F} = \nabla \times \mathbf{A}$ , then  $\mathbf{F}$  is certainly solenoidal because

$$\nabla \cdot (\nabla \times \mathbf{A}) = \frac{\partial}{\partial x_i} \varepsilon_{ijk} \frac{\partial A_k}{\partial x_j} = 0$$

Such an  $\mathbf{A}$  is called a *vector potential*. The converse also holds in a suitable domain  $\mathcal{D}$ .

**Remark.** For the converse to hold, we need  $\mathcal{D}$  to be simply connected, and we also need *surfaces* in  $\mathcal{D}$  to be able to be shrunk to a point. But this is complicated and non-examinable.

## §3 Orthogonal curvilinear coordinate systems

## §3.1 General curvilinear coordinate systems

**Definition.** A *curvilinear coordinate system* in 3D is a parameterisation  $(u, v, w)$  of  $\mathbb{R}^3$  – in other words, a surjective map  $\mathbf{x}(u, v, w)$  from a subset of  $\mathbb{R}^3$  to  $\mathbb{R}^3$ .

A well-known example in  $\mathbb{R}^2$  is plane polar coordinates.

**Remark.** Ideally, the map  $\mathbf{x}$  would be invertible, but in practice most coordinate systems fail to be invertible at points – for example, we can't invert at the origin in polar coordinates.

We define the *scale factors*

$$h_u = \left| \frac{\partial \mathbf{x}}{\partial u} \right|, \quad h_v = \left| \frac{\partial \mathbf{x}}{\partial v} \right|, \quad h_w = \left| \frac{\partial \mathbf{x}}{\partial w} \right|$$

and corresponding unit vectors

$$\mathbf{e}_u = \frac{1}{h_u} \frac{\partial \mathbf{x}}{\partial u}, \quad \mathbf{e}_v = \frac{1}{h_v} \frac{\partial \mathbf{x}}{\partial v}, \quad \mathbf{e}_w = \frac{1}{h_w} \frac{\partial \mathbf{x}}{\partial w}$$

**Definition.** An *orthogonal curvilinear coordinate system* is one in which  $\mathbf{e}_u$ ,  $\mathbf{e}_v$ ,  $\mathbf{e}_w$  are mutually orthogonal, and we normally re-order them to form a right-handed set. This allows them to be used as a basis.

Importantly, this basis is *local* – the basis vectors depend on position. For example, a vector field  $\mathbf{F}(\mathbf{x})$  can be expressed at  $\mathbf{x}$  in terms of its components relative to the basis vectors at  $\mathbf{x}$  – so  $\mathbf{F} = F_u \mathbf{e}_u + F_v \mathbf{e}_v + F_w \mathbf{e}_w$ . But even if  $\mathbf{F}$  is a constant vector, *its components won't be*.

By extending [proposition 1.3](#) to the vector function  $\mathbf{x}(u, v, w)$ , we have the following result:

**Proposition 3.1**

$$\begin{aligned} d\mathbf{x} &= \frac{\partial \mathbf{x}}{\partial u} du + \frac{\partial \mathbf{x}}{\partial v} dv + \frac{\partial \mathbf{x}}{\partial w} dw \\ &= h_u du \mathbf{e}_u + h_v dv \mathbf{e}_v + h_w dw \mathbf{e}_w \end{aligned}$$

whereas in Cartesian coordinates it would simply be  $du\mathbf{e}_u + dv\mathbf{e}_v + dw\mathbf{e}_w$ . Similarly, the position vector is *not* given by  $u\mathbf{e}_u + v\mathbf{e}_v + w\mathbf{e}_w$  – there is a difference between the coordinates  $(u, v, w)$  and the position vector  $\mathbf{x}(u, v, w)$ .

Consider a displacement that keeps  $u$  fixed, but allows  $v$  and  $w$  to vary. Then, from [proposition 3.1](#),  $d\mathbf{x}$  is a linear combination of  $\mathbf{e}_v$  and  $\mathbf{e}_w$ , so  $d\mathbf{x} \perp \mathbf{e}_u$  for all possible  $d\mathbf{x}$ . Hence  $\mathbf{e}_u$  is perpendicular to any surface of constant  $u$ , and similarly for  $\mathbf{e}_v$  and  $\mathbf{e}_w$ . So surfaces of constant  $u$ , constant  $v$  and constant  $w$  all intersect orthogonally at any given point.

**§3.2 Cylindrical polar coordinates**

**Definition.** Cylindrical polar coordinates are given by

$$\begin{aligned} x &= \rho \cos \phi \\ y &= \rho \sin \phi \\ z &= z \end{aligned}$$

DIAGRAM! where  $\rho \geq 0$  and  $\phi$  can be restricted to  $[0, 2\pi)$ , but doesn't need to be. We can invert this using

$$\begin{aligned} \rho &= \sqrt{x^2 + y^2} \\ \tan \phi &= y/x \\ z &= z \end{aligned}$$

Then surfaces of constant  $\rho$  are cylinders around the  $z$ -axis; of constant  $\phi$  are vertical planes through the  $z$ -axis; and of constant  $z$  are horizontal planes.

Different notations are used – notably  $r$  for  $\rho$  and  $\theta$  for  $\phi$ .

We have the following results about cylindrical polar coordinates:

**Proposition 3.2**

$$\begin{aligned} h_\rho &= 1 & h_\phi &= \rho & h_z &= 1 \\ \mathbf{e}_\rho &= \begin{pmatrix} \cos \phi \\ \sin \phi \\ 0 \end{pmatrix} & \mathbf{e}_\phi &= \begin{pmatrix} -\sin \phi \\ \cos \phi \\ 0 \end{pmatrix} & \mathbf{e}_z &= \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \end{aligned}$$

We can confirm that  $\{\mathbf{e}_\rho, \mathbf{e}_\phi, \mathbf{e}_z\}$  is orthonormal and right-handed. From [proposition 3.1](#) we obtain

$$d\mathbf{x} = d\rho \mathbf{e}_\rho + \rho d\phi \mathbf{e}_\phi + dz \mathbf{e}_z$$

and

**Proposition 3.3**

$$\mathbf{x} = \rho \mathbf{e}_\rho + z \mathbf{e}_z$$

with no  $\phi \mathbf{e}_\phi$  – because  $\mathbf{e}_\rho$ 's already got  $\phi$  in it.

**§3.3 Spherical polar coordinates**

**Definition.** *Spherical polar coordinates* are given by

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

where  $r \geq 0$ ,  $\theta \in [0, \pi]$ , and  $\phi$  can be restricted to  $[0, 2\pi)$  – DIAGRAM!! We can invert this using

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\tan \theta = \frac{\sqrt{x^2 + y^2}}{z}$$

$$\tan \phi = y/x$$

Then surfaces of constant  $r$  are spheres centred at the origin; of constant  $\theta$  are cones with apex at the origin; and of constant  $\phi$  are vertical planes through the  $z$ -axis.

**Remark.** Some people order the coordinates as  $(r, \phi, \theta)$  instead. Some swap the definitions of  $\phi$  and  $\theta$ . Some do both. Some replace  $\theta$  with  $\frac{\pi}{2} - \theta$ . This can be extremely confusing.

**Proposition 3.4**

For spherical polar coordinates,

$$h_r = 1$$

$$h_\theta = r$$

$$h_\phi = r \sin \theta$$

$$\mathbf{e}_r = \begin{pmatrix} \sin \theta \cos \phi \\ \sin \theta \sin \phi \\ \cos \theta \end{pmatrix}$$

$$\mathbf{e}_\theta = \begin{pmatrix} \cos \theta \cos \phi \\ \cos \theta \sin \phi \\ -\sin \theta \end{pmatrix}$$

$$\mathbf{e}_\phi = \begin{pmatrix} -\sin \phi \\ \cos \phi \\ 1 \end{pmatrix}$$

We also have

**Proposition 3.5**

$$\mathbf{x} = r \mathbf{e}_r$$

Finally, from [proposition 3.1](#) we obtain

$$d\mathbf{x} = dr \mathbf{e}_r + r d\theta \mathbf{e}_\theta + r \sin \theta d\phi \mathbf{e}_\phi$$

### §3.4 Vector differential operators in orthogonal curvilinear coordinate systems

#### §3.4.1 General orthogonal curvilinear coordinate systems

If we have a function expressed in terms of  $u, v, w$ , i.e. in arbitrary orthogonal curvilinear coordinates, we would like to be able to express  $\nabla f$  in terms of  $u, v, w$  as well. Observe firstly that

$$\nabla f \cdot \mathbf{e}_u = \frac{\partial f}{\partial x_i} \frac{1}{h_u} \frac{\partial x_i}{\partial u} = \frac{1}{h_u} \frac{\partial f}{\partial u}$$

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Since  $\{\mathbf{e}_u, \mathbf{e}_v, \mathbf{e}_w\}$  form an orthonormal set, doing these for each of the basis vectors will give us the components of  $\nabla f$  along each basis vector. We have

$$\nabla f = \frac{1}{h_u} \frac{\partial f}{\partial u} \mathbf{e}_u + \frac{1}{h_v} \frac{\partial f}{\partial v} \mathbf{e}_v + \frac{1}{h_w} \frac{\partial f}{\partial w} \mathbf{e}_w$$

and furthermore, we can take out the  $f$  to get

#### Proposition 3.6

$$\nabla = \frac{1}{h_u} \mathbf{e}_u \frac{\partial}{\partial u} + \frac{1}{h_v} \mathbf{e}_v \frac{\partial}{\partial v} + \frac{1}{h_w} \mathbf{e}_w \frac{\partial}{\partial w}$$

Calculating the divergence, curl and Laplacian of a vector field in general orthogonal curvilinear coordinates is extremely messy. There are a range of ways to do it – we will see the best method in section 6.5.

It is non-examinable that, in full generality,

$$\begin{aligned} \nabla \cdot \mathbf{F} &= \frac{1}{h_u h_v h_w} \left( \frac{\partial}{\partial u} (h_v h_w F_u) + \frac{\partial}{\partial v} (h_u h_w F_v) + \frac{\partial}{\partial w} (h_u h_v F_w) \right) \\ \nabla \times \mathbf{F} &= \frac{1}{h_u h_v h_w} \begin{vmatrix} h_u \mathbf{e}_u & h_v \mathbf{e}_v & h_w \mathbf{e}_w \\ \frac{\partial}{\partial u} & \frac{\partial}{\partial v} & \frac{\partial}{\partial w} \\ h_u F_u & h_v F_v & h_w F_w \end{vmatrix} \\ \nabla^2 f &= \frac{1}{h_u h_v h_w} \left( \frac{\partial}{\partial u} \left( \frac{h_v h_w}{h_u} \cdot \frac{\partial f}{\partial u} \right) + \frac{\partial}{\partial v} \left( \frac{h_u h_w}{h_v} \cdot \frac{\partial f}{\partial v} \right) + \frac{\partial}{\partial w} \left( \frac{h_u h_v}{h_w} \cdot \frac{\partial f}{\partial w} \right) \right) \end{aligned}$$

#### §3.4.2 Cylindrical polar coordinates

For a scalar field  $f(\rho, \phi, z)$  and vector field  $F_\rho \mathbf{e}_\rho + F_\phi \mathbf{e}_\phi + F_z \mathbf{e}_z$ , we obtain

$$\begin{aligned} \nabla f &= \frac{\partial f}{\partial \rho} \mathbf{e}_\rho + \frac{1}{\rho} \frac{\partial f}{\partial \phi} \mathbf{e}_\phi + \frac{\partial f}{\partial z} \mathbf{e}_z \\ \nabla \cdot \mathbf{F} &= \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho F_\rho) + \frac{1}{\rho} \frac{\partial F_\phi}{\partial \phi} + \frac{\partial F_z}{\partial z} \\ \nabla \times \mathbf{F} &= \left( \frac{1}{\rho} \frac{\partial F_z}{\partial \phi} - \frac{\partial F_\phi}{\partial z} \right) \mathbf{e}_\rho + \left( \frac{\partial F_\rho}{\partial z} - \frac{\partial F_z}{\partial \rho} \right) \mathbf{e}_\phi + \frac{1}{\rho} \left( \frac{\partial}{\partial \rho} (\rho F_\phi) - \frac{\partial F_\rho}{\partial \phi} \right) \mathbf{e}_z \end{aligned}$$

$$\nabla^2 f = \frac{\partial^2 f}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial f}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 f}{\partial \phi^2} + \frac{\partial^2 f}{\partial z^2}$$

It is required to know  $\nabla f$  and the radial terms of  $\nabla^2 f$  by heart. It is not necessary to learn div and curl.

Usefully, if we want to get the operators for plane polar coordinates, we can just throw out all terms involving  $z$ .

### §3.4.3 Spherical polar coordinates

For  $f(r, \theta, \phi)$  and  $F_r \mathbf{e}_r + F_\theta \mathbf{e}_\theta + F_\phi \mathbf{e}_\phi$ , we have

$$\begin{aligned} \nabla f &= \frac{\partial f}{\partial r} \mathbf{e}_r + \frac{1}{r} \frac{\partial f}{\partial \theta} \mathbf{e}_\theta + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} \mathbf{e}_\phi \\ \nabla \cdot \mathbf{F} &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 F_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (F_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial F_\phi}{\partial \phi} \\ \nabla \times \mathbf{F} &= \frac{1}{r \sin \theta} \left( \frac{\partial}{\partial \theta} (F_\phi \sin \theta) - \frac{\partial F_\theta}{\partial \phi} \right) \mathbf{e}_r + \\ &\quad \frac{1}{r} \left( \frac{1}{\sin \theta} \frac{\partial F_r}{\partial \phi} - \frac{\partial}{\partial r} (r F_\phi) \right) \mathbf{e}_\theta + \\ &\quad \frac{1}{r} \left( \frac{\partial}{\partial r} (r F_\theta) - \frac{\partial F_r}{\partial \theta} \right) \mathbf{e}_\phi \\ \nabla^2 f &= \frac{\partial^2 f}{\partial r^2} + \frac{2}{r} \frac{\partial f}{\partial r} + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left( \frac{\partial f}{\partial \theta} \sin \theta \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2} \end{aligned}$$

Again, it is required to know only  $\nabla f$  and the radial terms of  $\nabla^2 f$  by heart.

**Example** • Let  $f(\mathbf{x}) = |\mathbf{x}|^2$  and  $\mathbf{F}(\mathbf{x}) = \mathbf{x}$ . We have already obtained  $\nabla f = 2\mathbf{x}$  and  $\nabla \cdot \mathbf{F} = 3$  using Cartesian coordinates.

Instead working in cylindricals, we have  $\mathbf{x} = \rho\mathbf{e}_\rho + z\mathbf{e}_z$ , so  $f(\rho, \phi, z) = \rho^2 + z^2$ , while  $F_\rho = \rho$ ,  $F_\phi = 0$ ,  $F_z = z$ . Now we can apply the formulae:

$$\begin{aligned}\nabla f &= \frac{\partial f}{\partial \rho}\mathbf{e}_\rho + \frac{1}{\rho}\frac{\partial f}{\partial \phi}\mathbf{e}_\phi + \frac{\partial f}{\partial z}\mathbf{e}_z = 2\rho\mathbf{e}_\rho + 2z\mathbf{e}_z = 2\mathbf{x} \\ \nabla \cdot \mathbf{F} &= \frac{1}{\rho}\frac{\partial}{\partial \rho}(\rho F_\rho) + \frac{1}{\rho}\frac{\partial F_\phi}{\partial \phi} + \frac{\partial F_z}{\partial z} = 2 + 1 = 3\end{aligned}$$

- In sphericals, we have  $\mathbf{x} = r\mathbf{e}_r$ , so  $f(r, \theta, \phi) = r^2$ , while  $F_r = r$ ,  $F_\theta = 0$ ,  $F_\phi = 0$ . Hence

$$\begin{aligned}\nabla f &= \frac{\partial f}{\partial r}\mathbf{e}_r = 2r\mathbf{e}_r = 2\mathbf{x} \\ \nabla \cdot \mathbf{F} &= \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 F_r) = \frac{1}{r^2}\frac{\partial}{\partial r}(r^3) = 3\end{aligned}$$

- Consider the gravitational force field  $\mathbf{F} = -(GMm/r^2)\mathbf{e}_r$  of a fixed mass  $M$  at the origin acting on another mass  $m$ . We have

$$\nabla \cdot \mathbf{F} = -\frac{1}{r^2}\frac{\partial}{\partial r}(r^2 \cdot \frac{GMm}{r^2}) = 0$$

- The following simple example shows how we could obtain the expressions given above. Consider a simplified field of the form  $\mathbf{F} = \alpha(\theta)\mathbf{e}_\theta$ . We have

$$\begin{aligned}\nabla \times \mathbf{F} &= \left( \frac{\partial}{\partial r}\mathbf{e}_r + \frac{1}{r}\frac{\partial}{\partial \theta}\mathbf{e}_\theta + \frac{1}{r\sin\theta}\frac{\partial}{\partial \phi}\mathbf{e}_\phi \right) \times (\alpha(\theta)\mathbf{e}_\theta) \\ &= \mathbf{e}_r \times \alpha(\theta)\frac{\partial \mathbf{e}_\theta}{\partial r} + \frac{1}{r}\mathbf{e}_\theta \times \frac{\partial}{\partial \theta}(\alpha(\theta)\mathbf{e}_\theta) + \frac{1}{r\sin\theta}\mathbf{e}_\phi \times \alpha(\theta)\frac{\partial \mathbf{e}_\theta}{\partial \phi} \\ &= \dots\end{aligned}$$

## §4 Differential geometry of curves, surfaces and volumes

### §4.1 Curves and arc length

**Definition.** A *curve*  $C$  is a map  $\mathbf{x} : [a, b] \subset \mathbb{R} \rightarrow \mathbb{R}^3$ , in other words  $\mathbf{x}(t) = (x(t), y(t), z(t))$  for  $a \leq t \leq b$ . We say that the curve is *parameterised* by  $t$ . A curve is *closed* if  $\mathbf{x}(a) = \mathbf{x}(b)$ .

In this course we will require that curves be continuous and piecewise smooth (infinitely differentiable everywhere except a finite number of points). DIAGRAM: mostly smooth but a few spiky points.

Normally, we will use the word curve to refer to the *image* of the map  $C$ , not the map itself.

**Definition.** The *arc length* of a curve is given by

$$s = \int ds = \int |\dot{\mathbf{x}}(t)| dt$$

This is motivated by

$$\begin{aligned}\delta s = |\delta \mathbf{x}| &= \sqrt{\delta x^2 + \delta y^2 + \delta z^2} \\ &= \sqrt{\left(\frac{\delta x}{\delta t}\right)^2 + \left(\frac{\delta y}{\delta t}\right)^2 + \left(\frac{\delta z}{\delta t}\right)^2} \\ &= \left|\frac{\delta \mathbf{x}}{\delta t}\right| \delta t\end{aligned}$$

which, taking the limit, gives

### Proposition 4.1

$$\frac{ds}{dt} = |\dot{\mathbf{x}}|$$

Thus we obtain  $s$  as a function of  $t$  which we can, at least theoretically, invert to reparameterise a curve in terms of its arc length, which could be quite useful – the so-called *natural parameterisation*.

- Given two curves  $C_1$  and  $C_2$  with matching end-points, we define the curve  $C_1 + C_2$  to be the two curves joined end-to-end.
- The curve  $-C$  is the curve  $C$  traversed in the opposite direction –  $\mathbf{x}_{-C}(t) = \mathbf{x}_C(a + b - t)$ .

**Example** • Consider  $\mathbf{x}(t) = (at^2, 2at, 0)$ ,  $t \in [-1, 1]$ . This is a parameterisation (the standard one) of the parabola  $y^2 = 4ax$  between  $y = \pm 2a$ .

- $\mathbf{x}(t) = (a \cos t, b \sin t, 0)$ ,  $t \in [0, \pi]$  is a parameterisation of the upper half of the ellipse  $x^2/a^2 + y^2/b^2 = 1$ .
- $\mathbf{x}(t) = (\cos t, \sin t, t)$  is a helix. In cylindrical polars,  $\rho = 1$ ,  $\phi = t$ ,  $z = t$ , in other words the curve winds its way around the unit cylinder.

When drawing our curves, we should add an arrow to indicate the direction in which  $t$  increases.

### Example (Bonus non-examinable!)

Let  $R(t) = 2 + \cos \alpha t$  and  $\mathbf{x}(t) = (R(t) \cos \beta t, R(t) \sin \beta t, \sin \alpha t)$ . This curve winds its way around the torus given in cylindrical polars by  $(\rho - 2)^2 + z^2 = 1$ . If  $\alpha/\beta \in \mathbb{Q}$ , the curve closes on itself – in fact, if  $\alpha/\beta = p/q$  where  $p$  and  $q$  are coprime integers, then we have the *torus knot*  $K_{q,p}$ . But if  $\alpha/\beta \notin \mathbb{Q}$ , the curve covers the entire torus.

## §4.2 The Frenet-Serret equations

We will assume in this section that  $\mathbf{x}$  is parameterised by arc length  $s$ . If it is not, then the following results *do not apply*. For clarity, we will use primes to denote differentiation with respect to this canonical parameterisation. We define

**Proposition 4.2**

$$\begin{aligned} \mathbf{t} &= \mathbf{x}' && \text{the } \textit{tangent} \text{ vector} \\ \mathbf{n} &= \frac{1}{|\mathbf{x}''|} \mathbf{x}'' && \text{the } \textit{principal normal} \text{ vector} \\ \mathbf{b} &= \mathbf{t} \times \mathbf{n} && \text{the } \textit{binormal} \text{ vector} \end{aligned}$$

**Proposition 4.3** (Frenet-Serret equations)

$$\begin{aligned} \mathbf{t}' &= \kappa \mathbf{n} \\ \mathbf{n}' &= \tau \mathbf{b} - \kappa \mathbf{t} \\ \mathbf{b}' &= -\tau \mathbf{n} \end{aligned}$$

where  $\kappa = |\mathbf{x}''|$  and  $\tau = -\mathbf{b}' \cdot \mathbf{n}$  are scalars, not necessarily constant, called the *curvature* and *torsion* of the curve respectively.

*Proof.* First,  $ds = |d\mathbf{x}|$ , so  $|\frac{d\mathbf{x}}{ds}| = 1$  – hence  $|\mathbf{t}| = 1$ . Now  $\kappa = |\mathbf{x}''| = |\mathbf{t}'|$ , so  $\mathbf{n} = \mathbf{t}'/\kappa$  by definition. Hence  $|\mathbf{n}| = 1$  and  $\mathbf{t}' = \kappa \mathbf{n}$  (later on we'll worry about  $\kappa = 0$ ).

Differentiating  $\mathbf{t} \cdot \mathbf{t} = 1$  gives  $2\mathbf{t} \cdot \mathbf{t}' = 0$ , so  $\mathbf{t} \cdot \mathbf{n} = 0$ , that is  $\mathbf{t} \perp \mathbf{n}$ . Then, by the definition  $\mathbf{b} = \mathbf{t} \times \mathbf{n}$ , we can be sure  $\{\mathbf{t}, \mathbf{n}, \mathbf{b}\}$  is orthonormal and right-handed. This set (which depends on the position along the curve) is known as the *moving trihedral*.

Now  $\mathbf{b} \cdot \mathbf{b} = 1$ , so  $\mathbf{b} \cdot \mathbf{b}' = 0$ , and  $\mathbf{t} \cdot \mathbf{b} = 0$ , so

$$\begin{aligned} 0 &= \mathbf{t}' \cdot \mathbf{b} + \mathbf{t} \cdot \mathbf{b}' \\ &= \kappa \mathbf{n} \cdot \mathbf{b} + \mathbf{t} \cdot \mathbf{b}' \\ &= \mathbf{t} \cdot \mathbf{b}' \end{aligned}$$

Therefore  $\mathbf{b}' \perp \mathbf{t}$  and  $\mathbf{b}' \perp \mathbf{b}$ , so it must be that  $\mathbf{b}' \parallel \mathbf{n}$  and, from the definition of  $\tau$ ,  $\mathbf{b}' = -\tau \mathbf{n}$ .

Finally,  $\mathbf{n} = \mathbf{b} \times \mathbf{t}$ , so

$$\mathbf{n}' = \mathbf{b}' \times \mathbf{t} + \mathbf{b} \times \mathbf{t}' = \tau \mathbf{n} \times \mathbf{t} + \mathbf{b} \times \kappa \mathbf{n} = \tau \mathbf{b} - \kappa \mathbf{t}$$

as required.

Let's now worry about  $\kappa = 0$ . In that case,  $\mathbf{n}$ ,  $\mathbf{b}$  and  $\tau$  are also all undefined. If  $\kappa = 0$  everywhere, the curve is just a straight line. ■

**Definition.** The *radius of curvature* is defined as  $R = \kappa^{-1}$ .

At a point  $\mathbf{x}$ , the *osculating plane* is the plane that passes through  $\mathbf{x}$  and is spanned by  $\mathbf{t}$  and  $\mathbf{n}$ . Then a circle of radius  $R$  centred at  $\mathbf{x} + R\mathbf{n}$  is the best possible approximation of the curve at  $\mathbf{x}$  by a circle – it has the same values of  $\mathbf{x}$ ,  $\mathbf{x}'$  and  $\mathbf{x}''$  as the curve itself (as well as  $\kappa$ , of course).

DIAGRAM: diagram.

### §4.3 Using the Frenet-Serret equations with other parameterisations

One option is always to simply reparameterise our curve in terms of  $s$ . But this is not really necessary. Instead, we can simply apply the chain rule:

$$\begin{aligned}\mathbf{t} &= \mathbf{x}' = \frac{d\mathbf{x}}{dt} \frac{dt}{ds} = \frac{\dot{\mathbf{x}}}{|\dot{\mathbf{x}}|} \\ \mathbf{x}'' &= \frac{d\mathbf{t}}{ds} = \frac{d\mathbf{t}}{dt} \frac{dt}{ds} = \frac{\dot{\mathbf{t}}}{|\dot{\mathbf{x}}|}\end{aligned}$$

so that  $\kappa = \frac{|\dot{\mathbf{t}}|}{|\dot{\mathbf{x}}|}$ ,  $\mathbf{n} = \frac{\dot{\mathbf{t}}}{|\dot{\mathbf{t}}|}$ , and so on.

#### Example

Consider  $\mathbf{x}(t) = (\frac{1}{\sqrt{2}}t^2, t, t)$ . Then  $\dot{\mathbf{x}} = (\sqrt{2}t, 1, 1)$ , so  $\frac{ds}{dt} = |\dot{\mathbf{x}}| = \sqrt{2}\sqrt{1+t^2}$ . Hence

$$\begin{aligned}\mathbf{t} &= \mathbf{x}' = \frac{\dot{\mathbf{x}}}{|\dot{\mathbf{x}}|} = \frac{1}{\sqrt{2}\sqrt{1+t^2}}(\sqrt{2}t, 1, 1) \\ \dot{\mathbf{t}} &= \frac{1}{\sqrt{2}(1+t^2)^{3/2}}(\sqrt{2}, -t, -t) \\ \kappa &= |\mathbf{t}'| = \frac{|\dot{\mathbf{t}}|}{|\dot{\mathbf{x}}|} = \frac{1}{\sqrt{2}(1+t^2)^{3/2}}\end{aligned}$$

DIAGRAM: diagram

### §4.4 Surfaces, surface elements and the normal

**Definition.** A *surface* is a map  $\mathbf{x} : U \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^3$ . We typically use  $u$  and  $v$  to refer to the arguments of  $\mathbf{x}$ .

As with curves, we will normally refer to the image of the map, not the map itself.

#### Example

We might wish to make a hemispherical surface,  $H_R = \{\mathbf{x} : |\mathbf{x}| = R, z < 0\}$ . We can use spherical polar coordinates to parameterise this as

$$\mathbf{x}(\theta, \phi) = \begin{pmatrix} R \sin \theta \cos \phi \\ R \sin \theta \sin \phi \\ R \cos \theta \end{pmatrix}$$

with  $\theta \in (\pi/2, \pi]$ ,  $\phi \in [0, 2\pi)$ .

#### Lemma

Given a parallelogram with two sides  $\mathbf{a}$  and  $\mathbf{b}$ , its area is  $|\mathbf{a} \times \mathbf{b}|$ .

Consider now a region lying between curves of constant  $u$  and curves of constant  $v$ . What is the area of such a region?

Suppose we have two such curves meeting at  $\mathbf{x}(u, v)$ , and nearby curves at constant  $u + \delta u$  and  $v + \delta v$ . The part of  $S$  bounded by these curves forms, at lowest order, a very small parallelogram, with sides given by

$$\begin{aligned}\mathbf{x}(u + \delta u, v) - \mathbf{x}(u, v) &= \frac{\partial \mathbf{x}}{\partial u} \delta u \\ \mathbf{x}(u, v + \delta v) - \mathbf{x}(u, v) &= \frac{\partial \mathbf{x}}{\partial v} \delta v\end{aligned}$$

Hence, by the lemma, the area of the region is given by

#### Proposition 4.4

$$dS = \left| \frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v} \right| du dv$$

This infinitesimal parallelogram is referred to as a *surface element*.

**Remark.** Importantly, the normal to the surface at  $\mathbf{x}$  is parallel to  $\frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v}$ .

## §4.5 Volumes and volume elements

**Definition.** A *volume* is a map  $\mathbf{x} : U \subseteq \mathbb{R}^3 \rightarrow \mathbb{R}^3$ . We typically use  $u$ ,  $v$  and  $w$  to refer to the arguments of  $\mathbf{x}$ .

#### Example

To parameterise a sphere, we can do a similar thing to before, but let  $r \in [0, R]$ .

Similarly, we can divide our volumes into regions lying between *surfaces* of constant  $u$ ,  $v$  and  $w$ . To lowest order, these regions are *parallelepipeds* with sides  $\frac{\partial \mathbf{x}}{\partial u} \delta u$ ,  $\frac{\partial \mathbf{x}}{\partial v} \delta v$  and  $\frac{\partial \mathbf{x}}{\partial w} \delta w$ . Hence the volume element is given by

#### Proposition 4.5

$$dV = \left| \frac{\partial \mathbf{x}}{\partial u} \cdot \left( \frac{\partial \mathbf{x}}{\partial v} \times \frac{\partial \mathbf{x}}{\partial w} \right) \right| du dv dw$$

## §5 Line, area, surface and volume integrals

### §5.1 Line integrals

**Definition.** Given a curve  $C$  parameterised by  $t \in [a, b]$ , the *line integral* of a vector field  $\mathbf{F}$  along  $C$  is

$$\int_C \mathbf{F} \cdot d\mathbf{x} = \int_a^b \mathbf{F}(\mathbf{x}(t)) \cdot \frac{d\mathbf{x}}{dt} dt$$

An alternative but equivalent definition is to return to a Riemann sum: we define

$$\int_C \mathbf{F} \cdot d\mathbf{x} = \lim_{\Delta \rightarrow 0} \sum_{n=0}^{N-1} \mathbf{F}(\mathbf{x}_n) \cdot \delta \mathbf{x}_n$$

**Remark.** We can also take a line integral on a scalar field, or take a line integral while crossing (!)  $\mathbf{F}$  with  $d\mathbf{x}$  instead of dotting.

If  $C$  is a closed curve, then the line integral around the whole curve has a special notation

$$\oint_C \mathbf{F} \cdot d\mathbf{x}$$

and this is sometimes known as the *circulation* of  $\mathbf{F}$  around  $C$ . It also doesn't matter where we start along such a curve, so long as we go all the way around.

We have some elementary properties of line integrals:

- $\int_{-C} \mathbf{F} \cdot d\mathbf{x} = -\int_C \mathbf{F} \cdot d\mathbf{x}$ ;
- $\int_{C_1+C_2} \mathbf{F} \cdot d\mathbf{x} = \int_{C_1} \mathbf{F} \cdot d\mathbf{x} + \int_{C_2} \mathbf{F} \cdot d\mathbf{x}$ .

**Example** • Consider  $\mathbf{F} = (y, -x, 0)$ , and  $C$  the straight line joining  $(-1, 0, 0)$  to  $(1, 0, 0)$ . Parameterise the curve as  $\mathbf{x} = (t, 0, 0)$  for  $t \in [-1, 1]$ . Then  $\mathbf{F}(\mathbf{x}(t)) = (0, -t, 0)$ , and  $\dot{\mathbf{x}} = (1, 0, 0)$ , so  $\int_{C_1} \mathbf{F} \cdot d\mathbf{x} = 0$ .

- Consider  $\mathbf{F} = (3x^2, 2y, 0)$  on the same line. We obtain  $\mathbf{F} \cdot \dot{\mathbf{x}} = (3t^2, 0, 0) \cdot (1, 0, 0)$  and so  $\int_{C_1} \mathbf{F} \cdot d\mathbf{x} = \int_{-1}^1 3t^2 dt = 2$ .
- Consider  $\mathbf{F} = (y, -x, 0)$  along the curve  $C_2$  joining  $(-1, 0, 0)$  to  $(1, 0, 0)$  parameterised as  $\mathbf{x} = (t-1, t(t-2), t(t^2-4))$  for  $t \in [0, 2]$ . We have

$$\mathbf{F} = (t(t-2), 1-t, 0) \quad \mathbf{x} = (1, 2t-2, 3t^2-4)$$

so

$$\begin{aligned} \int_{C_2} \mathbf{F} \cdot d\mathbf{x} &= \int_0^2 t(t-2) - 2(t-1)^2 dt \\ &= -\frac{8}{3} \end{aligned}$$

- Consider  $\mathbf{F} = (3x^2, 2y, 0)$  along  $C_2$ . We have

$$\begin{aligned} \int_{C_2} \mathbf{F} \cdot d\mathbf{x} &= \int_0^2 (3(t-1)^2, 2t(t-2), 0) \cdot (1, 2t-2, 3t^2-4) dt \\ &= \int_0^2 4t^3 - 9t^2 + 2t - 3 dt \\ &= 2 \end{aligned}$$

- What is the work done by a force field  $\mathbf{F} = (y^2 e^x, e^y + 2ye^x, 0)$  moving a particle along  $C_3$ , which is the ellipse  $x^2/a^2 + y^2/b^2 = 1$  in the  $(x, y)$ -plane starting at  $(a, 0, 0)$  and going anticlockwise to  $(0, b, 0)$ ?

First, we need to parameterise the curve by  $\mathbf{x}(\theta) = (a \cos \theta, b \sin \theta, 0)$  for  $\theta \in [0, \pi/2]$ . Then

$$\mathbf{F}(\mathbf{x}(\theta)) = (b^2 \sin^2 \theta e^{a \cos \theta}, e^{b \sin \theta} + 2b \sin \theta e^{a \cos \theta}, 0)$$

so that

$$\mathbf{F} \cdot \frac{d\mathbf{x}}{d\theta} = -ab^2 \sin^3 \theta e^{a \cos \theta} + b \cos \theta e^{b \sin \theta} + 2b^2 \sin \theta \cos \theta e^{a \cos \theta}$$

so

$$\begin{aligned} \int_{C_3} \mathbf{F} \cdot d\mathbf{x} &= [e^{b \sin \theta}]_0^{\pi/2} + \int_0^{\pi/2} b^2 \sin \theta e^{a \cos \theta} (-a + 2 \cos \theta + a \cos^2 \theta) d\theta \\ &= e^b - 1 + \int_1^0 b^2 e^{au} (a^2 - 2u - au^2) du \quad (u = \cos \theta) \\ &= e^b - 1 + [b^2 e^{au} (1 - u^2)]_1^0 \\ &= e^b - 1 + b^2 \end{aligned}$$

Around the whole ellipse, starting at  $(a, 0, 0)$  and going anticlockwise all the way round, we would find  $\oint \mathbf{F} \cdot d\mathbf{x} = 0$ .

### Example

Let's do one in cylindrical polar coordinates! Consider  $\mathbf{F} = z\mathbf{e}_\phi$  on the path  $\rho = z^2$ ,  $\phi = \pi z$  from  $z = 0$  to  $z = 1$ . Here, we have

$$\begin{aligned} d\mathbf{x} &= d\rho\mathbf{e}_\rho + \rho d\phi\mathbf{e}_\phi + dz\mathbf{e}_z \\ &= (2z\mathbf{e}_\rho + \pi\rho\mathbf{e}_\phi + \mathbf{e}_z) dz \end{aligned}$$

and, on the curve, this gives  $\mathbf{F} \cdot d\mathbf{x} = \pi\rho z dz = \pi z^3 dz$ . Hence

$$\int \mathbf{F} \cdot d\mathbf{x} = \int_0^1 \pi z^3 dz = \pi/4$$

## §5.2 Path independence

**Definition** (Exact differential forms – reminder). We say that a differential form  $\mathbf{F} \cdot d\mathbf{x}$  is *exact* if there exists a single-valued scalar field  $f(\mathbf{x})$  such that  $\mathbf{F} \cdot d\mathbf{x} = df$  for all  $d\mathbf{x}$ .

From [proposition 2.2](#), we know that  $df = \nabla f \cdot d\mathbf{x}$  – so  $\mathbf{F} \cdot d\mathbf{x}$  is exact if and only if there exists a single-valued function  $f$  such that  $\mathbf{F} = \nabla f$ , i.e. if  $\mathbf{F}$  is a conservative vector field.

If  $\mathbf{F}$  is a conservative vector field, and  $C$  is a path from  $\mathbf{a}$  to  $\mathbf{b}$ , then

$$\int_C \mathbf{F} \cdot d\mathbf{x} = \int_C df = f(\mathbf{b}) - f(\mathbf{a})$$

and this is notably independent of the particular path. So we can, if we like, write

$$\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{F} \cdot d\mathbf{x}$$

In particular, we have the important result that, if  $C$  is a closed curve,

$$\oint_C \mathbf{F} \cdot d\mathbf{x} = 0$$

for any conservative  $\mathbf{F}$ , since  $\mathbf{b} = \mathbf{a}$  and  $f$  is single-valued.

Now we have the awesome result for vector fields  $\mathbf{F}$  in a simply-connected domain:

- ①  $\mathbf{F}$  is a conservative vector field;
- ② there exists a single-valued function  $f(\mathbf{x})$  such that  $\mathbf{F} = \nabla f$ ;
- ③  $\mathbf{F} \cdot d\mathbf{x}$  is an exact differential;
- ④  $\mathbf{F}$  is irrotational;
- ⑤  $\nabla \times \mathbf{F} = \mathbf{0}$ ;
- ⑥  $\frac{\partial F_i}{\partial x_j} = \frac{\partial F_j}{\partial x_i}$  for all  $i, j$ ;
- ⑦ for any  $\mathbf{a}, \mathbf{b}$  in the domain,  $\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{F} \cdot d\mathbf{x}$  is path-independent;
- ⑧  $\oint_C \mathbf{F} \cdot d\mathbf{x} = 0$  for all closed curves  $C$  in the domain.

To prove ⑧  $\implies$  ⑦ and ⑦  $\implies$  ② is non-trivial and we'll have to wait until section 6.4.

### Example

Now we can revisit some of our earlier examples with a new zest. Observe

- for the field  $\mathbf{F} = (y, -x, 0)$ , we have  $\nabla \times \mathbf{F} = (0, 0, -2) \neq \mathbf{0}$ , so  $\mathbf{F}$  is not conservative - indeed, we obtained different values for different paths.
- for the field  $\mathbf{F} = (3x^2, 2y, 0)$ , we have  $\nabla \times \mathbf{F} = \mathbf{0}$ , or  $\mathbf{F} = \nabla(x^3 + y^2)$ , and indeed we obtained the same answers for both paths.
- for the field  $\mathbf{F} = (y^2 e^x, e^y + 2y e^x, 0)$ , we have  $\mathbf{F} = \nabla(e^y + y^2 e^x)$ , so we can avoid all the integration and directly obtain

$$\int_{C_3} \mathbf{F} \cdot d\mathbf{x} = [e^y + y^2 e^x]_{(a,0,0)}^{(0,b,0)} = e^b + b^2 - 1$$

- for the field  $\mathbf{F} = 0\mathbf{e}_\rho + z\mathbf{e}_\phi + 0\mathbf{e}_z$ , we have  $\nabla \times (z\mathbf{e}_\phi) = -\mathbf{e}_\rho \neq \mathbf{0}$ . But in fact, even if  $\nabla \times \mathbf{F}$  had vanished, we still could not have concluded that  $\mathbf{F}$  is conservative, because  $\mathbf{e}_\phi$  is undefined on the  $z$ -axis and so the domain is not simply connected.

## §5.3 Area integrals

To integrate over a region  $A \subset \mathbb{R}^2$ , we cover  $A$  with a large number of small disjoint subsets  $\delta A_i$ . Then

$$\int_A f(x, y) dA = \lim_{\Delta \rightarrow 0} \sum_{i=1}^N f(x_i, y_i) \delta A_i$$

where in particular  $\Delta$  is the maximum linear dimension of any of the  $\delta A_i$ .

Informally, we can think of  $\int_A f(x, y) dA$  as the volume under the graph of  $z = f(x, y)$ , in analogy with the 1D case.

Note that  $A$  refers to the region, as well as its area: we have

$$A = \int_A 1 dA$$

The most obvious way to divide  $A$  up is with rectangles parallel to the  $x$  and  $y$  axes, each with area  $\delta x \delta y$ . Summing over a horizontal strip with a fixed value of  $y$ , with constant height  $\delta y$  and letting  $\delta x \rightarrow 0$ , we obtain

$$\left( \int_{X(y)} f(x, y) dx \right) \delta y$$

where, for each  $y$ ,  $X(y) = \{x : (x, y) \in A\}$ . We then sum these over the range of  $y$ , taking  $\delta y \rightarrow 0$ , to obtain

$$\int_{y_{\min}}^{y_{\max}} \left( \int_{X(y)} f(x, y) dx \right) dy$$

But we could have equally well done this the other way to obtain

$$\int_{x_{\min}}^{x_{\max}} \left( \int_{Y(x)} f(x, y) dy \right) dx$$

**Proposition (Fubini's theorem)**

For any reasonable function  $f$  over a reasonable  $A$ , both of these are equal to  $\int_A f(x, y) dA$ .

**Remark.** Only very basic conditions are needed. For example, if  $f$  is piecewise continuous and the boundary of  $A$  is a finite piecewise smooth curve, then Fubini's theorem holds.

Other notations for  $\int_A f(x, y) dA$  include  $\int_A f(x, y) dx dy$ ,  $\int_A f(\mathbf{x}) d^2 \mathbf{x}$ ,  $\iint_A f(x, y) dA$ ,  $\int_A f(x, y) dS$  and even  $\int dx \int dy f(x, y)$ .

### Example

Let us integrate  $f(x, y) = xy^2$  over  $A$ , a triangle with vertices at  $(0, 0)$ ,  $(1, 1)$  and  $(1, -1)$  (DIAGRAM). We have

$$\begin{aligned}\int_A f(x, y) dA &= \int_0^1 \int_{-x}^x xy^2 dy dx \\ &= \int_0^1 x \left[ \frac{1}{3} y^3 \right]_{-x}^x dx \\ &= \int_0^1 \frac{2}{3} x^4 dx = \frac{2}{15}\end{aligned}$$

Alternatively, we could have done

$$\begin{aligned}\int_A xy^2 dA &= \int_{-1}^1 \int_{|y|}^1 xy^2 dx dy \\ &= \int_{-1}^1 y^2 \left[ \frac{1}{2} x^2 \right]_{|y|}^1 dy \\ &= \int_{-1}^1 \frac{1}{2} y^2 (1 - y^2) dy \\ &= \frac{2}{15}\end{aligned}$$

Importantly, the inside integral should have limits specified in terms of  $x$  or  $y$ .

In the special case that  $A$  is a rectangle  $[a, b] \times [c, d]$  and  $f$  is separable, i.e.  $f(x, y) = g(x)h(y)$ , then

$$\int_A f(x, y) dA = \left( \int_a^b g(x) dx \right) \left( \int_c^d h(y) dy \right)$$

## §5.4 Change of variables for area integrals

### Proposition

Let  $(u, v) \mapsto (x, y)$  be a bijection from a region  $A'$  in the  $(u, v)$ -plane to a region  $A$  in the  $(x, y)$ -plane. Then

$$\int_A f(x, y) dx dy = \int_{A'} F(u, v) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv$$

where  $F(u, v) = f(x(u, v), y(u, v))$  and

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \det \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix}$$

is called the *Jacobian* of the transformation. Importantly, it is the *modulus* of the Jacobian that appears in the integral.

**Remark.** We often call capital  $F$  little  $f$  instead, although this is an obvious abuse of notation.

**Remark.** The change of variables may fail to be a bijection for one or two points – c.f. polar coordinates. But this is ok.

*Proof.* Let us divide  $A$  up using curves of constant  $u$  and  $v$ . The area of each region is given by

$$\begin{aligned} \left| \frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v} \right| du dv &= \left| \begin{pmatrix} \frac{\partial x}{\partial u} \\ \frac{\partial y}{\partial u} \\ 0 \end{pmatrix} \times \begin{pmatrix} \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial v} \\ 0 \end{pmatrix} \right| du dv \\ &= \left| \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \frac{\partial y}{\partial u} \right| du dv \\ &= |J| du dv \end{aligned}$$

Then we simply need to add up all the infinitesimal areas  $f(\mathbf{x}(u, v))|J| du dv$ . ■

**Remark.** A full proof of this result requires first a proof of the cross product expression for the infinitesimal area.

### Example

Consider  $\int_A y^2 dA$  where  $A$  is the semi-circular domain  $\{(x, y) : x^2 + y^2 < R^2, y > 0\}$ . We choose plane polar coordinates  $(r, \theta)$  with  $x = r \cos \theta$ ,  $y = r \sin \theta$ . Then, we have a bijection between  $A' = \{(r, \theta) : r < R, 0 < \theta < \pi\}$ , so

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \det \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{pmatrix} = \det \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix} = r$$

and so

$$\begin{aligned} \int_A y^2 dA &= \int_{A'} r^2 \sin^2 \theta |r| dr d\theta \\ &= \int_0^R r^3 dr \int_0^\pi \sin^2 \theta d\theta \\ &= \frac{1}{8} \pi R^4 \end{aligned}$$

DIAGRAM of the area element.

## §5.5 Properties of the Jacobian

Suppose that  $J_1$  is the Jacobian matrix for the transformation  $(u, v) \mapsto (x, y)$ , and  $J_2$  is the Jacobian matrix for  $(\xi, \eta) \mapsto (u, v)$ . Then, consider the product

$$J_1 J_2 = \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} \begin{pmatrix} \frac{\partial u}{\partial \xi} & \frac{\partial u}{\partial \eta} \\ \frac{\partial v}{\partial \xi} & \frac{\partial v}{\partial \eta} \end{pmatrix}$$

$$\begin{aligned}
&= \begin{pmatrix} \frac{\partial x}{\partial u} \frac{\partial u}{\partial \xi} + \frac{\partial x}{\partial v} \frac{\partial v}{\partial \xi} & \frac{\partial x}{\partial u} \frac{\partial u}{\partial \theta} + \frac{\partial x}{\partial v} \frac{\partial v}{\partial \theta} \\ \frac{\partial y}{\partial u} \frac{\partial u}{\partial \xi} + \frac{\partial y}{\partial v} \frac{\partial v}{\partial \xi} & \frac{\partial y}{\partial u} \frac{\partial u}{\partial \theta} + \frac{\partial y}{\partial v} \frac{\partial v}{\partial \theta} \end{pmatrix} \\
&= \begin{pmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial \xi} & \frac{\partial y}{\partial \theta} \end{pmatrix}
\end{aligned}$$

In other words, the Jacobian of a composition of maps is the product of the Jacobians:

$$\frac{\partial(x, y)}{\partial(u, v)} \cdot \frac{\partial(u, v)}{\partial(\xi, \eta)} = \frac{\partial(x, y)}{\partial(\xi, \eta)}$$

A particularly nice consequence of this fact is that

$$\frac{\partial(x, y)}{\partial(u, v)} = \left( \frac{\partial(u, v)}{\partial(x, y)} \right)^{-1}$$

### Example

Let us calculate  $\int_A x^3 y \, dA$  where  $A$  is the area between the hyperbolas  $x^2 - y^2 = 1$  and  $x^2 - y^2 = 2$ , enclosed within the circle  $x^2 + y^2 = 4$ , with  $x, y > 0$ .

desperate need for a DIAGRAM.

Let us change variables to  $u = x^2 + y^2$ ,  $v = x^2 - y^2$ . Then

$$\frac{\partial(u, v)}{\partial(x, y)} = \det \begin{pmatrix} 2x & 2y \\ 2x & -2y \end{pmatrix} = -8xy$$

so  $\frac{\partial(x, y)}{\partial(u, v)} = -\frac{1}{8xy}$ . Now we must find the corresponding area, which we do by considering the boundary curves. The left and right boundaries of  $A$  correspond to  $v = 1$ ,  $1 < u < 4$  and  $v = 2$ ,  $2 < u < 4$  respectively. The lower boundary has  $y = 0$ , so  $u = v$  with  $1 < u < 2$ , and the upper boundary has  $u = 4$  with  $1 < v < 2$ . So

$$A' = \{(u, v) : 1 < v < 2, v < u < 4\}$$

Hence

$$\begin{aligned}
\int_A x^3 y \, dA &= \int_{A'} x^3 y \left| -\frac{1}{8xy} \right| \, du \, dv \\
&= \frac{1}{8} \int_{A'} x^2 \, du \, dv \\
&= \frac{1}{8} \int_{A'} \frac{1}{2}(u + v) \, du \, dv \\
&= \frac{1}{16} \int_1^2 \int_v^4 u + v \, du \, dv \\
&= \frac{1}{16} \int_1^2 8 + 4v - \frac{3}{2}v^2 \, dv
\end{aligned}$$

## §5.6 Volume integrals

Volume integrals are really the same as area integrals, just in one more dimension (in fact these notions will generalise to  $\mathbb{R}^n$ ). We cover  $V$  with a large number of small disjoint

volumes, and define

$$\int_V f(\mathbf{x}) dV = \lim_{\Delta \rightarrow 0} \sum_{i=1}^N f(\mathbf{x}_i) \delta V_i$$

where  $\Delta$  is the maximum linear dimension of all the  $\delta V_i$ .

Alternatively, we have the iterated integral definition, in particular

$$\int_{x_{\min}}^{x_{\max}} \left( \int_{Y(x)} \left( \int_{Z(x,y)} f(x, y, z) dz \right) dy \right) dx$$

Notations include  $\int_V f(\mathbf{x}) dV$ ,  $\int_V f(x, y, z) dx dy dz$ ,  $\iiint_V f(x, y, z) dV$ ,  $\int_V f(\mathbf{x}) d^3\mathbf{x}$  and  $\int dx \int dy \int dz f(x, y, z)$ .

### Example

Let  $V$  be the cuboid  $x \in [-a, a]$ ,  $y \in [-b, b]$ ,  $z \in [-c, c]$ . Let us calculate  $\int_V \mathbf{F} dV$  where  $\mathbf{F}$  is the vector field  $(z^2, 0, yz)$ .

Then, applying some separation,

$$\begin{aligned} \int_V \mathbf{F} dV &= \left( \int_V z^2 dV, 0, \int_V yz dV \right) \\ &= \left( \int_{-a}^a 1 dx \int_{-b}^b 1 dy \int_{-c}^c z^2 dz, 0, \int_{-a}^a 1 dx \int_{-b}^b y dy \int_{-c}^c z dz \right) \\ &= \left( \frac{8}{3} abc^3, 0, 0 \right) \end{aligned}$$

**Remark.** Sometimes, symmetries are quotable, like with  $yz$  as above.

## §5.7 Change of variables for volume integrals

### Proposition

Let  $(u, v) \mapsto (x, y)$  be a bijection from a region  $A'$  in the  $(u, v)$ -plane to a region  $A$  in the  $(x, y)$ -plane. Then

$$\int_A f(x, y, z) dx dy dz = \int_{A'} F(u, v, w) \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du dv dw$$

where  $F(u, v, w) = f(x(u, v, w), y(u, v, w), z(u, v, w))$  and

$$J = \frac{\partial(x, y, z)}{\partial(u, v, w)} = \det \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{pmatrix}$$

is the Jacobian.

*Proof.* The proof is almost identical. We have

$$dV = \left| \frac{\partial \mathbf{x}}{\partial u} \cdot \left( \frac{\partial \mathbf{x}}{\partial v} \times \frac{\partial \mathbf{x}}{\partial w} \right) \right| du dv dw = \left| \det \left( \frac{\partial \mathbf{x}}{\partial u} \left| \frac{\partial \mathbf{x}}{\partial v} \right| \frac{\partial \mathbf{x}}{\partial w} \right) \right| = |J| du dv dw$$

■

## Lecture 12 – 18/02/26

All properties of the Jacobian established in two variables also hold for three variables.

The most common changes of variables are to either cylindrical or spherical polar coordinates. We have

$$J_{\text{cyl}} = \det \begin{pmatrix} \cos \phi & -\rho \sin \phi & 0 \\ \sin \phi & \rho \cos \phi & 0 \\ 0 & 0 & 1 \end{pmatrix} = \rho$$

$$J_{\text{sph}} = \det \begin{pmatrix} \sin \theta \cos \phi & r \cos \theta \cos \phi & -r \sin \theta \sin \phi \\ \sin \theta \sin \phi & r \cos \theta \sin \phi & r \sin \theta \cos \phi \\ \cos \theta & -r \sin \theta & 0 \end{pmatrix} = r^2 \sin \theta$$

Furthermore,  $\rho$  and  $r^2 \sin \theta$  are always positive (recall  $\theta \in [0, \pi]$ ), so we don't need to worry about the absolute value.

### Example

Let us calculate the volume of the sphere  $r \leq R$ . We have

$$\begin{aligned} \int_{r \leq R} 1 dV &= \int_{r=0}^R \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} 1 \cdot r^2 \sin \theta dr d\theta d\phi \\ &= \int_0^R r^2 dr \int_0^{\pi} \sin \theta d\theta \int_0^{2\pi} d\phi \\ &= \frac{4}{3} \pi R^3 \end{aligned}$$

### Example

That was easy! Let's instead calculate the volume of a sphere with a cylinder of radius  $a$  along the  $z$ -axis removed from it. We should use cylindrical polars with the cylinder occupying  $\rho \leq a$  and the sphere  $\rho^2 + z^2 \leq R^2$ . We obtain

$$\begin{aligned} V &= \int_{\rho=a}^R \int_{\phi=0}^{2\pi} \int_{z=-\sqrt{R^2-\rho^2}}^{\sqrt{R^2-\rho^2}} 1 \cdot \rho dz d\phi d\rho \\ &= 2\pi \int_a^R 2\rho \sqrt{R^2 - \rho^2} d\rho \\ &= \frac{4}{3} \pi (R^2 - a^2)^{3/2} \end{aligned}$$

## §5.8 Classification of surfaces

A surface  $S$  can be categorised as either *closed* or *open*. An open surface is one which has a boundary curve, denoted  $\partial S$ . The surface is said to *span* the curve. A *closed* surface is one with no boundary curve –  $\partial S = \emptyset$ .

### Example

The hemispherical shell  $H_R = \{\mathbf{x} : |\mathbf{x}| = R, z < 0\}$  is an open surface with  $\partial H_R = \{\mathbf{x} : x^2 + y^2 = R^2, z = 0\}$ , i.e. a circle.

The spherical shell  $S_R = \{\mathbf{x} : |\mathbf{x}| = R\}$  is a closed surface, as it has no boundary.

We can further categorise surfaces as *orientable* (having two sides) or *non-orientable* (having only one side). We shall only consider orientable surfaces.

The important thing about closed orientable surfaces is that they have an inside and an outside, while an open or non-orientable surface does not.

These definitions also only apply to finite surfaces – whenever we are considering surfaces of infinite extent, we will treat them as the limit of finite surfaces.

## §5.9 Surface integrals

Integrating over a two-dimensional surface  $S$  embedded in  $\mathbb{R}^3$  is unsurprisingly similar to an area integral. We divide  $S$  into small subsets  $\delta S_i$ , and define

$$\int_S f(\mathbf{x}) dS = \lim_{\Delta \rightarrow 0} \sum_{i=1}^n f(\mathbf{x}_i) \delta S_i$$

Importantly, these  $\delta S_i$  are not flat, unlike in the case of an area integral. Alternative notations include  $\iint_S f(\mathbf{X}) dS$  and, if  $S$  is closed,  $\oint_S f(\mathbf{x}) dS$ .

At each point on an orientable surface there will be two unit normals, pointing in opposite directions. Either normal can be denoted  $\hat{\mathbf{n}}$  – the one we choose depends on the particular application.

If  $\mathbf{F}$  is a vector field, we might want to consider  $\int_S \mathbf{F}(\mathbf{x}) \cdot \hat{\mathbf{n}}(\mathbf{x}) dS$ . This is known as the *flux* through  $S$ . We also define

$$d\mathbf{S} = \hat{\mathbf{n}} dS$$

so that the flux of  $\mathbf{F}$  through  $S$  is

$$\int_S \mathbf{F} \cdot d\mathbf{S}$$

### Proposition 5.1 (The magic formula)

If  $S$  is parameterised by  $u$  and  $v$ , the vector  $d\mathbf{S}$  is given by

$$d\mathbf{S} = \pm \frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v} du dv$$

*Proof.* Using [proposition 4.4](#), we have

$$\hat{\mathbf{n}} = \pm \frac{\frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v}}{\left| \frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v} \right|}, \quad dS = \left| \frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v} \right| du dv$$

from which the result follows. ■

Using these results reduces a surface integral to an area integral in the flat  $(u, v)$ -plane.

### Example

Let us find the surface area of the hemispherical shell  $H_R = \{\mathbf{x} : |\mathbf{x}| = R, z < 0\}$ . Let us parameterise  $H_R$  by

$$\mathbf{x}(\theta, \phi) = \begin{pmatrix} R \sin \theta \cos \phi \\ R \sin \theta \sin \phi \\ R \cos \theta \end{pmatrix}$$

with  $\theta \in (\frac{\pi}{2}, \pi]$ ,  $\phi \in [0, 2\pi)$ . Then

$$\frac{\partial \mathbf{x}}{\partial \theta} \times \frac{\partial \mathbf{x}}{\partial \phi} = R \begin{pmatrix} \cos \theta \cos \phi \\ \cos \theta \sin \phi \\ -\sin \theta \end{pmatrix} \times R \begin{pmatrix} -\sin \theta \sin \phi \\ \sin \theta \cos \phi \\ 0 \end{pmatrix} = R^2 \begin{pmatrix} \sin^2 \theta \cos \phi \\ \sin^2 \theta \sin \phi \\ \sin \theta \cos \theta \end{pmatrix}$$

so

$$dS = R^2 \sin \theta \, d\theta \, d\phi$$

Hence, the surface area is

$$\begin{aligned} \int_{H_R} 1 \, dS &= \int_{\theta=\frac{\pi}{2}}^{\pi} \int_{\phi=0}^{2\pi} R^2 \sin \theta \, d\theta \, d\phi \\ &= R^2 \int_{\frac{\pi}{2}}^{\pi} \sin \theta \, d\theta \int_0^{2\pi} 1 \, d\phi \\ &= 2\pi R^2 \end{aligned}$$

**Example**

Let us find the flux of  $\mathbf{F}(\mathbf{x}) = \frac{\alpha}{|\mathbf{x}|^3} \mathbf{x}$ , with constant  $\alpha$ , through the surface of a sphere  $S_R = \{\mathbf{x} : |\mathbf{x}| = R\}$ , where our normal points away from the origin.

We will use the same parameterisation, but this time allow  $\theta \in [0, \pi]$ . Recalling our expression for  $\frac{\partial \mathbf{x}}{\partial \theta} \times \frac{\partial \mathbf{x}}{\partial \phi}$  from above, and picking the positive sign to point away from the origin, we have

$$\mathbf{F} \cdot d\mathbf{S} = \frac{\alpha}{R^3} \begin{pmatrix} R \sin \theta \cos \phi \\ R \sin \theta \sin \phi \\ R \cos \theta \end{pmatrix} \cdot R^2 \sin \theta \begin{pmatrix} \sin^2 \theta \cos \phi \\ \sin^2 \theta \sin \phi \\ \sin \theta \cos \theta \end{pmatrix} = \alpha \sin \theta \, d\theta d\phi$$

Now it merely remains to do

$$\begin{aligned} \int_{S_R} \mathbf{F} \cdot d\mathbf{S} &= \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \alpha \sin \theta \, d\theta d\phi \\ &= \alpha \int_{\theta=0}^{\pi} \sin \theta \, d\theta \int_{\phi=0}^{2\pi} 1 \, d\phi \\ &= 4\pi\alpha \end{aligned}$$

**Remark.** Since  $d\mathbf{S}$  is a normal vector pointing out with magnitude equal to a surface area element, we can rationalise that

$$d\mathbf{S} = R^2 \sin \theta \, \mathbf{e}_r \, d\theta d\phi$$

for a sphere of radius  $R$ . Likewise, in cylindrical coordinates for a cylinder with radius  $a$ , the vector surface element is

$$d\mathbf{S} = a \mathbf{e}_\rho \, d\phi dz$$

**Example**

Let us find the flux of  $\mathbf{F}(\mathbf{x}) = (y^2 e^{-(y^2+z^2)}, e^{-y^4}, -x^2 e^{-(y^2+z^2)})$  through the infinite plane  $\Pi = \{\mathbf{x} : x - z = 0\}$ , where we choose the normal pointing in the positive  $x$ -direction (DIAGRAM).

We can parameterise the plane using  $x$  and  $y$ , such that  $\mathbf{x} = (x, y, x)$ . Then

$$\frac{\partial \mathbf{x}}{\partial x} \times \frac{\partial \mathbf{x}}{\partial y} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

Note, however, that here we do need a negative sign, so  $d\mathbf{S} = (1, 0, -1) dx dy$ . Hence the flux is

$$\begin{aligned} \int_{\Pi} \mathbf{F} \cdot d\mathbf{S} &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x^2 + y^2) e^{-(x^2+y^2)} dx dy & (*) \\ &= \int_{-\infty}^{\infty} x^2 e^{-x^2} dx \int_{-\infty}^{\infty} e^{-y^2} dy + \int_{-\infty}^{\infty} e^{-x^2} dx \int_{-\infty}^{\infty} y^2 e^{-y^2} dy \\ &= 2 \cdot \frac{\sqrt{\pi}}{2} \cdot \sqrt{\pi} = \pi \end{aligned}$$

**Remark.** We could have alternatively found  $(*)$  by transforming to plane polars.

**Remark.** We could have easily worked out the direction of  $d\mathbf{S}$  just by thinking about the geometry a little bit. However, the magic formula is important for finding the area of the surface element.

## §6 Integral theorems

In general, integral theorems aim to reduce the dimension of an integral, much as the fundamental theorem of calculus does for normal integrals.

### §6.1 Green's theorem in the plane

#### Theorem 6.1 (Green's theorem in the plane)

Given an area  $A \subset \mathbb{R}^2$  with piecewise smooth boundary  $C = \partial A$  and continuously differentiable functions  $P(x, y)$  and  $Q(x, y)$ ,

$$\int_A \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \oint_C (P dx + Q dy)$$

where  $C$  is traversed in the positive sense (normally meaning anti-clockwise).

*Proof.* Initially, suppose that  $A$  is a simple shape such that, for each  $y$ , there is a single range of values for  $x$ ,  $x^-(y) \leq x \leq x^+(y)$ ; likewise, for each  $x$ ,  $y^-(x) \leq y \leq y^+(x)$ . Then, we may divide  $C$  into  $C^+$  and  $C^-$ , which satisfy respectively  $x = x^+(y)$  and  $x = x^-(y)$

for every  $y$  in the range. Now

$$\begin{aligned}\int_A \frac{\partial Q}{\partial x} dx dy &= \int_{y_{\min}}^{y_{\max}} \int_{x^-(y)}^{x^+(y)} \frac{\partial Q}{\partial x} dx dy \\ &= \int_{y_{\min}}^{y_{\max}} Q(x^+(y), y) - Q(x^-(y), y) dy \\ &= \int_{C^+} Q(x, y) dy + \int_{C^-} Q(x, y) dy \\ &= \oint_C Q dy\end{aligned}$$

Note that the minus sign is lost since  $C^-$  goes from  $y_{\max}$  to  $y_{\min}$ , unlike  $C^+$ . The exact same approach will work for the  $\frac{\partial P}{\partial y}$  integral with some care, which will suffice for the result.

This completes the proof of Green's theorem for simple shapes. If, on the other hand,  $A$  is a 'complicated' shape, we claim we can break it up into simple shapes  $A_i$ , and apply Green's theorem to each of these. ■

**Remark.** In particular, we note that  $\int_A \frac{\partial Q}{\partial x} dx dy = \sum_i \int_{A_i} \frac{\partial Q}{\partial x} dx dy$ , since the  $A_i$  are disjoint. Furthermore,  $\oint_{\partial A} Q dy = \sum_i \oint_{\partial A_i} Q dy$ , because, although each  $\partial A_i$  contains segments not part of  $\partial A$ , those contributions exactly cancel out. Hence the result follows.

## Lecture 14 – 23/02/26

### Example

Let us consider the ellipse  $x^2/a^2 + y^2/b^2 \leq 1$  with  $P = -\frac{1}{2}y$ ,  $Q = \frac{1}{2}x$ . Then

$$\pi ab = \int_A 1 dx dy = \int_A \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

and

$$\begin{aligned}\oint_C P dx + Q dy &= \int_0^{2\pi} \left( -\frac{1}{2}b \sin t, \frac{1}{2}a \cos t \right) \cdot (-a \sin t, b \cos t) dt \\ &= \pi ab\end{aligned}$$

### §6.1.1 Multiple boundary curves

We may also consider areas with multiple boundary curves – in particular, internal boundary curves as well as external boundary curves. In this case, it is necessary to traverse internal boundaries *clockwise* as opposed to counter-clockwise.

To illustrate this, let us consider an area  $A$  with outer boundary  $C_1$  traversed counter-clockwise and one inner boundary  $C_2$  traversed clockwise.

Let  $A_2$  be the interior of  $C_2$ , and  $A_1 = A \cup A_2$ , so  $A_1$  is the entire interior of  $C_1$ . Then  $\int_{A_1} = \int_A + \int_{A_2}$ , so  $\int_A = \int_{A_1} - \int_{A_2} = \oint_{C_1} - \oint_{-C_2}$ , by Green's theorem – in particular, we integrate over  $-C_2$  since  $C_2$  is clockwise. Then clearly  $\int_A = \oint_{C_1} + \oint_{C_2}$  as required.

**Remark.** This argument unfortunately assumes that the integrand is defined in  $A_1$  (the area in the hole), which is of course not true in general. In the general case, we can put a cut in  $A$  between the interior and exterior boundaries – in that case, it's clear that, for the outer boundary to be traversed counter-clockwise, the inner boundary is traversed clockwise. Then, taking the cut arbitrarily small, the result follows.

## §6.2 Stokes' theorem

### Theorem 6.2 (Stokes' theorem)

If  $\mathbf{F}(x)$  is a continuously differentiable vector field and  $S$  is a piecewise smooth orientable open surface with piecewise smooth boundary curve  $C = \partial S$ ,

$$\int_S \nabla \times \mathbf{F} \cdot d\mathbf{S} = \oint_C \mathbf{F} \cdot d\mathbf{x}$$

where  $C$  is traversed in the positive sense with regard to the normal on  $S$ .

**Remark.** Note that Green's theorem in the plane is a 2D case of Stokes' theorem. Let  $S$  be a flat surface with  $z = 0$  and take  $\mathbf{F} = (P, Q, 0)$  – the result follows.

What does ‘the positive sense with regard to the normal on  $S$ ’ mean? Firstly, note that, since  $S$  is open, we must specify a normal to give any notion of direction. Then, we say that if  $\mathbf{t}$  is the tangent vector to the curve  $C$  and  $\mathbf{n}$  is the normal to  $S$  at the same surface, then  $\mathbf{t} \times \mathbf{n}$  should point away from the surface. Equivalently, if one walks along  $C$  with one's feet on  $C$  and one's head in the direction of  $N$ , the surface should be on one's left. DIAGRAM!

Alternatively, one has the right-hand rule – if one's thumb points along  $\hat{\mathbf{n}}$ , then one's curled fingers indicate the direction of  $C$ .

*Sketch proof, non-examinable.* We start from Green's theorem, parameterising the surface  $S$  using  $(u, v) \in D \subset \mathbb{R}^2$ . Then the right hand side becomes

$$\oint_C \mathbf{F} \cdot d\mathbf{x} = \oint_{\partial D} \mathbf{F} \cdot \left( \frac{\partial \mathbf{x}}{\partial u} du + \frac{\partial \mathbf{x}}{\partial v} dv \right)$$

Letting  $P(u, v) = \mathbf{F}(\mathbf{x}(u, v)) \cdot \frac{\partial \mathbf{x}}{\partial u}$  and  $Q(u, v) = \mathbf{F}(\mathbf{x}(u, v)) \cdot \frac{\partial \mathbf{x}}{\partial v}$  and apply Green's theorem in the  $(u, v)$ -plane to obtain

$$\int_D \left( \frac{\partial Q}{\partial u} - \frac{\partial P}{\partial v} \right) du dv$$

Then, one can use suffix notation and the chain rule to show

$$(\nabla \times \mathbf{F}) \cdot \left( \frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v} \right) = \frac{\partial}{\partial u} \left( \mathbf{F} \cdot \frac{\partial \mathbf{x}}{\partial v} \right) - \frac{\partial}{\partial v} \left( \mathbf{F} \cdot \frac{\partial \mathbf{x}}{\partial u} \right)$$

from which the result follows. ■

**Example**

Consider the open surface  $S$  given by  $z = (x^2 + y^2)^2$ ,  $0 \leq z \leq 1$  and the vector field  $\mathbf{F} = (x^2y, 0, 0)$ . We use cylindrical polars, so  $S$  becomes  $z = \rho^4$ , so that  $S$  is parameterised by  $\mathbf{x} = (\rho \cos \phi, \rho \sin \phi, \rho^4)$ . Applying the magic formula,

$$d\mathbf{S} = - \begin{pmatrix} \cos \phi \\ \sin \phi \\ 4\rho^3 \end{pmatrix} \times \begin{pmatrix} -\rho \sin \phi \\ \rho \cos \phi \\ 0 \end{pmatrix} d\rho d\phi = \begin{pmatrix} 4\rho^4 \cos \phi \\ 4\rho^4 \sin \phi \\ -\rho \end{pmatrix} d\rho d\phi$$

where we have chosen the negative sign to get the downward-facing normal. Now  $\nabla \times \mathbf{F} = (0, 0, -x^2) = (0, 0, -\rho^2 \cos^2 \phi)$ , so

$$\int_S \nabla \times \mathbf{F} \cdot d\mathbf{S} = \int_{\rho=0}^1 \int_{\phi=0}^{2\pi} \rho^3 \cos^2 \phi d\phi d\rho = \frac{\pi}{4}$$

Now for the line integral.  $C = \partial S$  is the curve  $x^2 + y^2 = 1$ ,  $z = 1$ , that is  $\rho = 1$ ,  $z = 1$ ,  $\phi \in [0, 2\pi)$ . To determine the direction, apply the right hand rule – we see that  $C$  must be traversed in the direction of decreasing  $\phi$ , or clockwise when viewed from above. Hence we have

$$\begin{aligned} \oint_C \mathbf{F} \cdot d\mathbf{x} &= \int_{2\pi}^0 x^2 y \frac{d\mathbf{x}}{d\phi} d\phi \\ &= \int_0^{2\pi} \cos^2 \phi \sin^2 \phi d\phi \end{aligned}$$

**§6.2.1 Closed surfaces**

Stokes' theorem also applies to closed surfaces  $S$ , which have  $\partial S = \emptyset$ . Hence

$$\oint_S \nabla \times \mathbf{F} \cdot d\mathbf{S} = 0$$

**§6.2.2 Multiple boundary curves**

We may again have the case of two or more boundary curves. We have to apply the right hand rule for the directions of  $\mathbf{n}$  and the curve separately for *each* boundary.

### Example (hard)

We are asked to verify Stokes' theorem for  $\mathbf{F}(\mathbf{x}) = (y, z, x^2 + y^2)$  on the hyperboloid of one sheet  $z^2 + 1 = x^2 + 4y^2$  for  $0 \leq z \leq \sqrt{3}$ .

We choose the normal pointing away from the axis, and parameterise the surface by

$$\mathbf{x}(z, \alpha) = \left( \sqrt{z^2 + 1} \cos \alpha, \frac{1}{2} \sqrt{z^2 + 1} \sin \alpha, z \right)$$

Why not cylindrical coordinates? Well, we don't have circular symmetry anymore, so we should adjust to slightly scaled cylindrical polar coordinates. Applying the magic formula,

$$d\mathbf{S} = \pm \begin{pmatrix} \frac{z}{\sqrt{z^2+1}} \cos \alpha \\ \frac{z}{\sqrt{z^2+1}} \sin \alpha \\ 1 \end{pmatrix} \times \begin{pmatrix} -\sqrt{z^2+1} \sin \alpha \\ \frac{1}{2} \sqrt{z^2+1} \cos \alpha \\ 0 \end{pmatrix} dz d\alpha = \pm \begin{pmatrix} -\frac{1}{2} \sqrt{z^2+1} \cos \alpha \\ -\sqrt{z^2+1} \sin \alpha \\ \frac{1}{2} z \end{pmatrix} dz d\alpha$$

We choose the normal with the minus sign. On the surface,  $\nabla \times \mathbf{F} = (2y - 1, -2x, -1) = (\sqrt{z^2 + 1} \sin \alpha - 1, -2\sqrt{z^2 + 1} \cos \alpha, -1)$ , so

$$\begin{aligned} \int \nabla \times \mathbf{F} \cdot d\mathbf{S} &= \int_{z=0}^{\sqrt{3}} \int_{\alpha=0}^{2\pi} \left( -\frac{3}{2}(z^2 + 1) \sin \alpha \cos \alpha - \frac{1}{2} \sqrt{z^2 + 1} \cos \alpha + \frac{1}{2} z \right) d\alpha dz \\ &= \pi \int_0^{\sqrt{3}} z dz = \frac{3\pi}{2} \end{aligned}$$

Now we consider the boundary curves, which are

$$\begin{aligned} C_1 : x^2 + 4y^2 &= 1, \quad z = 0 \\ C_2 : x^2 + 4y^2 &= 4, \quad z = \sqrt{3} \end{aligned}$$

On  $z = \sqrt{3}$ , the right-hand rule shows that we need to move in the direction of decreasing  $\phi$ . Hence, we take the parameterisation  $\mathbf{x}(t) = (2 \cos t, \sin t, \sqrt{3})$  with  $\mathbf{F} = (\sin t, \sqrt{3}, 4 \cos^2 t + \sin^2 t)$  for  $t$  decreasing from  $2\pi$  to  $0$ . Hence

$$\int_{C_2} \mathbf{F} \cdot d\mathbf{x} = \int_{2\pi}^0 (-2 \sin^2 t + \sqrt{3} \cos t) dt = 2\pi$$

On  $z = 0$ , we instead need to move in the direction of increasing  $\phi$ . We thus parameterise  $C_1$  as  $\mathbf{x}(t) = (\cos t, \frac{1}{2} \sin t, 0)$  with  $\mathbf{F} = (\frac{1}{2} \sin t, 0, \cos^2 t + \frac{1}{2} \sin^2 t)$  for  $t$  increasing from  $0$  to  $\pi$ . Hence

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{x} = \int_0^{\pi} \left( -\frac{1}{2} \sin^2 t \right) dt = -\frac{\pi}{2}$$

To verify Stokes' theorem, we observe

$$\frac{3\pi}{2} = 2\pi - \frac{\pi}{2}$$

### §6.3 The divergence theorem

#### Theorem 6.3 (Divergence theorem)

Let  $\mathbf{F}(\mathbf{x})$  be a continuously differentiable vector field and  $V$  a volume enclosed by a piecewise smooth orientable *closed* surface  $S = \partial V$ . Then

$$\int_V \nabla \cdot \mathbf{F} \, dV = \int_S \mathbf{F} \cdot d\mathbf{S}$$

where the normal to  $S$  points outwards from  $V$ .

The theorem is also known as Gauss's theorem, or sometimes Green's theorem.

*Sketch proof, non-examinable.* To prove this we once again start by assuming  $V$  is a simple shape so that, for each fixed  $y$  and  $z$ , we have a single range of  $x$ , that is  $x^-(y, z) < x < x^+(y, z)$  (DIAGRAM).  $S$  can then be divided into a right-hand part  $S^+$  on which  $x = x^+(y, z)$  and a left-hand part  $S^-$  on which  $x = x^-(y, z)$ . Now

$$\begin{aligned} \int_V \frac{\partial F_1}{\partial x} \, dV &= \int_z \int_y \left( \int_{x^-(y,z)}^{x^+(y,z)} \frac{\partial F_1}{\partial x} \, dx \right) dy \, dz \\ &= \int_z \int_y (F_1(x^+(y, z), y, z) - F_1(x^-(y, z), y, z)) \, dy \, dz \\ &= \int_{S^+} F_1(\mathbf{x}) \, dy \, dz - \int_{S^-} F_1(\mathbf{x}) \, dy \, dz \end{aligned}$$

We parameterise  $S^+$  using  $y$  and  $z$ , so  $\mathbf{x} = (x^+(y, z), y, z)$  on  $S^+$ , so

$$d\mathbf{S} = \pm \begin{pmatrix} \frac{\partial x^+}{\partial y} \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} \frac{\partial x^+}{\partial z} \\ 0 \\ 1 \end{pmatrix} dy \, dz = \pm \begin{pmatrix} 1 \\ -\frac{\partial x^+}{\partial y} \\ -\frac{\partial x^+}{\partial z} \end{pmatrix} dy \, dz$$

and therefore  $\mathbf{e}_1 \cdot d\mathbf{S} = \pm dy \, dz$ . Since we have chosen the outward normal, we want the plus sign. Similarly, on  $S^-$ ,  $\mathbf{e}_1 \cdot d\mathbf{S} = -dy \, dz$ , so

$$\int_V \frac{\partial F_1}{\partial x} \, dV = \int_{S^+} F_1 \mathbf{e}_1 \cdot d\mathbf{S} + \int_{S^-} F_1 \mathbf{e}_1 \cdot d\mathbf{S} = \int_S F_1 \mathbf{e}_1 \cdot d\mathbf{S}$$

Analogous calculations can clearly be carried out for  $\int_V \frac{\partial F_2}{\partial y} \, dV$  and  $\int_V \frac{\partial F_3}{\partial z} \, dV$ , yielding  $\int_S F_2 \mathbf{e}_2 \cdot d\mathbf{S}$  and  $\int_S F_3 \mathbf{e}_3 \cdot d\mathbf{S}$  respectively. Addition yields the result.

For more complicated shapes, we split the shape up as in the proof of Green's theorem in the plane. ■

### Example

Let us calculate the flux of the vector field  $\mathbf{F} = (y, x, z^6)$  through the surface of the sphere  $V_R = \{\mathbf{x} : |\mathbf{x}| < R\}$ ? By the divergence theorem, it is  $\int_{V_R} 6z^5 dV$ , which vanishes by oddness in  $z$ .

But that's no fun. Let us calculate the flux directly. We've already calculated the surface element for a sphere, namely

$$d\mathbf{S} = R^2 \sin \theta \begin{pmatrix} \sin \theta \cos \phi \\ \sin \theta \sin \phi \\ \cos \theta \end{pmatrix} d\theta d\phi = R^2 \sin \theta \mathbf{e}_r d\theta d\phi$$

so that

$$\begin{aligned} \int_{\partial V_R} \mathbf{F} \cdot d\mathbf{S} &= \int_{\partial V_R} \begin{pmatrix} R \sin \theta \sin \phi \\ R \sin \theta \cos \phi \\ R^6 \cos^6 \theta \end{pmatrix} \cdot R^2 \sin \theta \begin{pmatrix} \sin \theta \cos \phi \\ \sin \theta \sin \phi \\ \cos \theta \end{pmatrix} d\theta d\phi \\ &= \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} (2R^3 \sin^2 \theta \sin \phi \cos \phi + R^8 \cos^7 \theta) \sin \theta d\theta d\phi \end{aligned}$$

Both terms vanish by oddness. Specifically, we have  $\int_0^{2\pi} \sin \phi \cos \phi d\phi = 0$  (the range can be replaced with  $\int_{-\pi}^{\pi}$  and the integrand is odd) and  $\int_0^{\pi} \cos^7 \theta \sin \theta d\theta = 0$  (because the integrand is odd around  $\frac{\pi}{2}$ ). Hence the divergence theorem is confirmed.

**Remark.** In such integrals, one should be on the lookout for integral tricks. For example,

$$\begin{aligned} \int_0^{2\pi} \sin^{2m+1} \phi d\phi &= \int_0^{2\pi} \cos^{2m+1} \phi d\phi = 0 \\ \int_0^{2\pi} \sin^{2m+1} \phi \cos^n \phi d\phi &= \int_0^{2\pi} \sin^m \phi \cos^{2n+1} \phi d\phi = 0 \\ \int_0^{\pi} \sin^m \theta \cos^{2n+1} \theta d\theta &= 0 \end{aligned}$$

because they're all odd about something or other.

### Example (hard)

Let us find the flux of  $\mathbf{F}(\mathbf{x}) = \mathbf{x}$  through the surface

$$y^2 + z^2 = \beta^2(x+1)^2$$

where  $\beta$  is a positive constant,  $1 \leq x \leq 2$ , and the normal points away from the  $x$ -axis (DIAGRAM). Afterwards, we will confirm the result using the divergence theorem.

$S$  is in fact an open cone with apex at  $(-1, 0, 0)$  and the  $x$ -axis as its axis of symmetry. We would like to use cylindrical polars, but rotated such that

$$x = x, \quad y = \rho \cos \phi, \quad z = \rho \sin \phi$$

We then parameterise the cone with

$$\mathbf{x} = (x, \beta(x+1) \cos \phi, \beta(x+1) \sin \phi)$$

Hence

$$d\mathbf{S} = \pm \begin{pmatrix} 1 \\ \beta \cos \phi \\ \beta \sin \phi \end{pmatrix} \times \begin{pmatrix} 0 \\ -\beta(x+1) \sin \phi \\ \beta(x+1) \cos \phi \end{pmatrix} dx d\phi = \pm \begin{pmatrix} \beta^2(x+1) \\ -\beta(x+1) \cos \phi \\ -\beta(x+1) \sin \phi \end{pmatrix} dx d\phi$$

We need the minus sign to ensure that  $d\mathbf{S}$  points away from the  $x$ -axis – see diagram. So the flux is given by

$$\begin{aligned} \mathcal{F}_S &= \int_{x=1}^2 \int_{\phi=0}^{2\pi} \begin{pmatrix} x \\ \beta(x+1) \cos \phi \\ \beta(x+1) \sin \phi \end{pmatrix} \cdot \begin{pmatrix} -\beta^2(x+1) \\ \beta(x+1) \cos \phi \\ \beta(x+1) \sin \phi \end{pmatrix} dx d\phi \\ &= \int_{x=1}^2 \int_{\phi=0}^{2\pi} \beta^2(x+1) dx d\phi = 5\pi\beta^2 \end{aligned}$$

To apply the divergence theorem, it is necessary to turn  $S$  into a closed surface, which we will do with plane caps at  $x = 1$  and  $x = 2$ . Calling the enclosed volume  $V$ , we note that the normals point outwards as hoped. Let the fluxes through the cap at  $x = 1$  and  $x = 2$  be  $\mathcal{F}_1$  and  $\mathcal{F}_2$  respectively. According to the divergence theorem, then,

$$\int_V \nabla \cdot \mathbf{F} dV = \mathcal{F}_S + \mathcal{F}_1 + \mathcal{F}_2 \implies \mathcal{F}_S = \int_V 3 dV - \mathcal{F}_1 - \mathcal{F}_2$$

Then

$$\begin{aligned} \int_V 3 dV &= \int_{x=1}^2 \int_{\rho=0}^{\beta(x+1)} \int_{\phi=0}^{2\pi} 3\rho d\phi d\rho dx = 19\pi\beta^2 \\ \mathcal{F}_1 &= \int_{\substack{x=1 \\ y^2+z^2 \leq 4\beta^2}} \begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix} dy dz = - \int_{y^2+z^2 \leq 4\beta^2} dy dz = -\pi(2\beta)^2 \\ \mathcal{F}_2 &= \int_{\substack{x=2 \\ y^2+z^2 \leq 9\beta^2}} \begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} dy dz = \int_{y^2+z^2 \leq 9\beta^2} 2 dy dz = 2\pi(3\beta)^2 \end{aligned}$$

and so  $\mathcal{F}_S = \pi\beta^2(19 + 4 - 18) = 5\pi\beta^2$  as required.

### §6.3.1 Multiple boundary surfaces

If  $V$  contains one or more holes, then  $\partial V$  will consist of two or more boundary surfaces. Since we need the normal to point outwards from  $V$ , it will point *into* the holes.

### Example

Let us apply the divergence theorem to the spherical annulus  $V = \{x \in \mathbb{R}^3 : a < |\mathbf{x}| < b\}$ .  $\partial V$  consists of two parts,  $|\mathbf{x}| = a$  and  $|\mathbf{x}| = b$  and we need  $d\mathbf{S} = \mathbf{e}_r dS$  on  $|\mathbf{x}| = b$ ,  $d\mathbf{S} = -\mathbf{e}_r dS$  on  $|\mathbf{x}| = a$ . Hence

$$\int_V \nabla \cdot \mathbf{F} dV = \int_{|\mathbf{x}|=b} \mathbf{F}(\mathbf{x}) \cdot \mathbf{e}_r dS - \int_{|\mathbf{x}|=a} \mathbf{F}(\mathbf{x}) \cdot \mathbf{e}_r dS$$

This is extremely useful if  $\mathbf{F}$  is undefined in the holes.

**Remark.** If we have an outer surface  $S_1$  and an inner surface  $S_2$ , some people will write

$$\int_V \nabla \cdot \mathbf{F} dV = \int_{S_1} \mathbf{F} \cdot d\mathbf{S} + \int_{S_2} \mathbf{F} \cdot d\mathbf{S}$$

where the understanding is that the normal for  $S_2$  is in the opposite direction. Other people will write

$$\int_V \nabla \cdot \mathbf{F} dV = \int_{S_1} \mathbf{F} \cdot d\mathbf{S} - \int_{S_2} \mathbf{F} \cdot d\mathbf{S}$$

with the understanding that the normals are in the same direction for both surfaces.

### §6.3.2 Connections to other integral theorems (non-examinable)

Green's theorem in the plane is a special case in 2D of the divergence theorem. Using  $\mathbf{F} = (Q, -P, 0)$ , the left hand side of [theorem 6.1](#) is  $\int_A \nabla \cdot \mathbf{F} dA$ . The right hand side is  $\int_C \mathbf{F} \cdot \mathbf{n} ds$  where  $ds$  is the infinitesimal arc length (the equivalent of  $dS$  in 1D), because  $\mathbf{n} ds = (dy, -dx, 0)$  is perpendicular to  $d\mathbf{x} = (dx, dy, 0)$ , has the correct magnitude, and is outwards from  $A$ .

In fact, these are all the same results in the differential geometry of manifolds, namely

$$\int_{\Omega} d\omega = \int_{\partial\Omega} \omega$$

where  $\omega$  is a differential form,  $d\omega$  its exterior derivative, and  $\Omega$  an orientable manifold. This is called the generalised Stokes' theorem, or sometimes just Stokes' theorem!

### §6.4 Scalar potentials for irrotational vector fields

We are now finally able to show that an irrotational vector field in a simply connected domain is conservative.

*Proof.* Firstly, observe that, if  $\oint_C \mathbf{F} \cdot d\mathbf{x} = 0$  for all closed curves  $C$ , then  $\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{F} \cdot d\mathbf{x}$  is path-independent because, if we consider two possible paths  $C_1$  and  $C_2$  from  $\mathbf{a}$  to  $\mathbf{b}$ , then  $C = C_1 + -C_2$  is a closed curve, and hence  $\int_{C_1} \mathbf{F} \cdot d\mathbf{x} - \int_{C_2} \mathbf{F} \cdot d\mathbf{x} = 0$  as required.

Now, given an irrotational vector field  $\mathbf{F}$  and an arbitrary closed curve  $C$ , let  $S$  be a surface that spans  $C$ , i.e.  $C = \partial S$ . By Stokes' theorem,  $\oint_C \mathbf{F} \cdot d\mathbf{x} = \int_S \nabla \times \mathbf{F} \cdot d\mathbf{S} = 0$ .

Now we define

$$f(\mathbf{x}) = \int_{\mathbf{x}_0}^{\mathbf{x}} \mathbf{F} \cdot d\mathbf{x}$$

From the above, this integral is path-independent and  $f$  is therefore well-defined (note  $\mathbf{x}_0$  is an arbitrary point within the domain). Now consider

$$f(\mathbf{x} + \delta\mathbf{x}) - f(\mathbf{x}) = \int_{\mathbf{x}}^{\mathbf{x} + \delta\mathbf{x}} \mathbf{F} \cdot d\mathbf{x} = \mathbf{F}(\mathbf{x}) \cdot \delta\mathbf{x} + O(|\delta\mathbf{x}|^2)$$

Hence  $df = f(\mathbf{x} + d\mathbf{x}) - f(\mathbf{x}) = \mathbf{F} \cdot d\mathbf{x}$ , which means that  $\nabla f = \mathbf{F}$ . This completes the proof. ■

**Remark.** But where did we use simply-connectedness? It was in the existence of  $S$ . If the domain  $\mathcal{D}$  is not simply-connected, then there are some closed curves  $C$  that cannot be spanned by a surface lying entirely in  $\mathcal{D}$ . If it is simply-connected, then we can define  $C_s$  for  $s \in [0, 1]$  with  $C_0 = C$ ,  $C_1$  a point, every  $C_s$  in  $\mathcal{D}$  and  $C_s$  continuous in  $s$ . Then  $S = \cup_{s \in [0, 1]} C_s$ .

## §6.5 Coordinate-independent definitions of vector differential operators (non-examinable)

Recall that we defined grad with respect to a standard basis, but also in a coordinate-independent way –  $df = \nabla f \cdot d\mathbf{x}$ . However, we did not do this for div and curl. But we can now rectify this!

Consider the divergence theorem applied to a very small volume  $V_\varepsilon$  of maximum linear dimension  $\varepsilon$ , containing the point  $\mathbf{x}$  with surface  $\partial V_\varepsilon$ . Then

$$\int_{V_\varepsilon} \nabla \cdot \mathbf{F} dV = (\nabla \cdot \mathbf{F})|_{\mathbf{x}} V_\varepsilon + \dots$$

where  $V_\varepsilon$  denotes both the volume and the volume's volume. Hence

$$\nabla \cdot \mathbf{F} = \lim_{\varepsilon \rightarrow 0} \frac{1}{V_\varepsilon} \int_{\partial V_\varepsilon} \mathbf{F} \cdot d\mathbf{S}$$

which can be used as the definition of  $\nabla \cdot \mathbf{F}$ .

For curl, let us take a fixed vector  $\mathbf{k}$ , and consider a small open surface  $S_\varepsilon(\mathbf{k})$  of maximum linear dimension  $\varepsilon$  that contains  $\mathbf{x}$  and has a normal  $\mathbf{n} = \mathbf{k}$  at  $\mathbf{x}$ . Then

$$\int_{S_\varepsilon(\mathbf{x})} \nabla \times \mathbf{F} \cdot d\mathbf{S} = (\nabla \times \mathbf{F})|_{\mathbf{x}} \cdot \mathbf{k} S_\varepsilon(\mathbf{k}) + \dots$$

Now we can just take  $\mathbf{k}$  to be each basis vector in turn, and hence calculate every component of  $\nabla \times \mathbf{F}$  in any coordinate system we like.

These definitions provide an efficient way to calculate the results for div and curl in general orthogonal curvilinear coordinate systems.

## §6.6 Green's identities and other corollaries

### Proposition (Green's first identity)

Let  $\phi$  and  $\psi$  be scalar fields defined in a closed volume  $V$  with surface  $\partial V$  and outward normal  $\mathbf{n}$ . Then

$$\int_V (\phi \nabla^2 \psi + \nabla \phi \cdot \nabla \psi) dV = \int_{\partial V} \phi \nabla \psi \cdot d\mathbf{S}$$

*Sketch proof.* We apply the divergence theorem to  $\mathbf{F} = \phi \nabla \psi$ , using the product rule to evaluate  $\nabla \cdot (\phi \nabla \psi)$ . ■

**Proposition 6.4** (Green's second identity)

Let  $\phi$  and  $\psi$  be scalar fields defined in a closed volume  $V$  with surface  $\partial V$  and outward normal  $\mathbf{n}$ . Then

$$\int_V (\phi \nabla^2 \psi - \psi \nabla^2 \phi) dV = \int_{\partial V} (\phi \nabla \psi - \psi \nabla \phi) \cdot d\mathbf{S}$$

*Proof.* An exercise. ■

Note that, in these equations, terms of the form  $\frac{\partial f}{\partial \mathbf{n}}$  (directional derivatives) appear. There is a special notation for this, in particular

$$\frac{\partial f}{\partial \mathbf{n}} = \nabla f \cdot \mathbf{n}$$

Then, for example, [proposition 6.4](#) becomes

$$\int_V (\phi \nabla^2 \psi - \psi \nabla^2 \phi) dV = \int_{\partial V} \left( \phi \frac{\partial \psi}{\partial \mathbf{n}} - \psi \frac{\partial \phi}{\partial \mathbf{n}} \right) dS$$

**Proposition 6.5**

For a scalar field  $f$ ,

$$\int_V \nabla f dV = \int_{\partial V} f d\mathbf{S}$$

*Proof.* Consider the divergence theorem applied to the vector field  $\mathbf{F} = f\mathbf{a}$  where  $f$  is a scalar field and  $\mathbf{a}$  is an arbitrary *constant* vector. Then  $\nabla \cdot \mathbf{F} = f \nabla \cdot \mathbf{a} + \mathbf{a} \cdot \nabla f$ , but  $\nabla \cdot \mathbf{a} = 0$ , so

$$\begin{aligned} \int_V \mathbf{a} \cdot \nabla f dV &= \int_{\partial V} f \mathbf{a} \cdot d\mathbf{S} \\ \implies \mathbf{a} \cdot \int_V \nabla f dV &= \mathbf{a} \cdot \int_{\partial V} f d\mathbf{S} \end{aligned}$$

Since  $\mathbf{a}$  is arbitrary, we may ‘cancel’. ■

**Example**

Consider  $f = xyz$  in the cube  $V$  with  $0 \leq x, y, z \leq 2$ . Then  $\nabla f = (yz, xz, xy)$ , so

$$\int_V \nabla f \, dV = \begin{pmatrix} \int_V yz \, dV \\ \int_V xz \, dV \\ \int_V xy \, dV \end{pmatrix} = \begin{pmatrix} 8 \\ 8 \\ 8 \end{pmatrix}$$

To confirm [proposition 6.5](#), we will calculate  $\int_{\partial} f \, dS$  by considering each of the surfaces. On the top side,  $z = 2$ ,  $f = 2xy$ ,  $\mathbf{n} = (0, 0, 1)$ , and  $dS = dx \, dy$ , so

$$\int_{\text{top}} f \, dS = \int_0^2 \int_0^2 2xy \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} dx \, dy = \begin{pmatrix} 0 \\ 0 \\ 8 \end{pmatrix}$$

On the bottom side, we will get  $\mathbf{0}$ , since  $z = 0$  – each of the sides along an axis will then give  $\mathbf{0}$ , while the others will give an 8 in each of the other directions.

**§6.7 Maxwell's equations and the continuity equation****Proposition 6.6** (Maxwell's equations)

Maxwell's equations in vacuum for the electric field  $\mathbf{E}(\mathbf{x}, t)$  and magnetic field  $\mathbf{B}(\mathbf{x}, t)$  in the presence of an electric charge density field  $\rho(\mathbf{x}, t)$  (charge per unit volume) and electric current density  $\mathbf{J}(\mathbf{x}, t)$  (current per unit area) are:

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \rho / \varepsilon_0 && \text{(Gauss's law)} \\ \nabla \cdot \mathbf{B} &= 0 && \text{(Gauss's law for magnetism)} \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} && \text{(Faraday's law)} \\ \nabla \times \mathbf{B} &= \mu_0 \left( \mathbf{J} + \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) && \text{(Ampère-Maxwell law)} \end{aligned}$$

It is not necessary to know these equations by heart.

**Proposition** (Gauss's law)

Let  $V$  be an arbitrary fixed volume containing total charge  $Q_V$ . Then

$$\int_{\partial V} \mathbf{E} \cdot d\mathbf{S} = \frac{1}{\varepsilon_0} Q_V$$

*Proof.* We have

$$Q_V = \int_V \rho \, dV = \int_V \varepsilon_0 \nabla \cdot \mathbf{E} \, dV$$

by Maxwell 1. Applying the divergence theorem yields the result. ■

**Remark.** One may note that we have called two different things Gauss's law. In fact, the two statements are called the *differential* and *integral* forms of the law, respectively, for obvious reasons.

**Remark.** On the other hand, we clearly obtain  $\int_{\partial V} \mathbf{B} \cdot d\mathbf{S} = 0$  from Gauss's law for magnetism – this is a statement of the non-existence of magnetic monopoles.

**Remark** (Non-examinable). A whole family of identities can be derived in similar ways. For example, starting from Faraday's law,

$$\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{S} = \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S} = - \int_S \nabla \times \mathbf{E} \cdot d\mathbf{S} = - \oint_{\partial S} \mathbf{E} \cdot d\mathbf{x}$$

by Stokes' theorem. In other words, changing magnetic flux causes circulation of the electric field.

Similarly, starting from Ampère-Maxwell, one obtains

$$\frac{d}{dt} \int_S \mathbf{E} \cdot d\mathbf{S} = \frac{1}{\varepsilon_0 \mu_0} \oint_{\partial S} \mathbf{B} \cdot d\mathbf{x} - \frac{1}{\varepsilon_0} \int_S \mathbf{J} \cdot d\mathbf{S}$$

– for constant  $\mathbf{E}$ , electric current causes circulation of the magnetic field.

### §6.7.1 Continuity equations

**Proposition 6.7** (Continuity equation for electric charge)

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0$$

*Proof.* Taking the divergence of Ampère-Maxwell and using the fact that  $\nabla \cdot (\nabla \times \mathbf{F}) = 0$ ,

$$0 = \nabla \cdot \mathbf{J} + \varepsilon_0 \nabla \cdot \frac{\partial \mathbf{E}}{\partial t} = \nabla \cdot \mathbf{J} + \varepsilon_0 \frac{\partial}{\partial t} (\nabla \cdot \mathbf{E})$$

by commutativity of  $\frac{\partial}{\partial t}$  and  $\frac{\partial}{\partial x_i}$ . Then, applying Gauss's law gives the result. ■

Integrating the continuity equation over a fixed volume  $V$ , we obtain

$$\begin{aligned} \int_V \frac{\partial \rho}{\partial t} dV &= - \int_V \nabla \cdot \mathbf{J} dV \\ \iff \frac{dQ_V}{dt} &= - \int_{\partial V} \mathbf{J} \cdot d\mathbf{S} \end{aligned}$$

in other words, the rate at which the charge in  $V$  decreases is equal to the net flux of current out of  $V$ . This represents *conservation of charge* – charge only changes when currents carry it into or out of the volume. In this context, we call  $\mathbf{J}$  the *flux of charge* (strictly speaking, the flux density of charge) – dotting it with  $d\mathbf{S}$  just extracts the component perpendicular to the surface, i.e. how much is actually moving in or out of  $V$ .

Similar continuity equations arise in many other areas of physics. For example, in fluid dynamics,

**Proposition 6.8** (Continuity equation for mass)

Consider a fluid with density  $\rho(\mathbf{x}, t)$  and velocity  $\mathbf{u}(\mathbf{x}, t)$  at some point. Then

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$

As before, this is equivalent to the statement that, for a volume  $V$

$$\frac{dm}{dt} = - \int_{\partial V} \rho \mathbf{u} \cdot d\mathbf{S}$$

so  $\rho \mathbf{u}$  is clearly the flux (density) of mass, and this equation is a statement of conservation of mass.

In the case of a fluid with constant density, [proposition 6.8](#) reduces to  $\nabla \cdot \mathbf{u} = 0$ , i.e.  $\mathbf{u}$  is an incompressible vector field.

## §7 Laplace's and Poisson's equations

### §7.1 Poisson's equation and its boundary conditions

**Definition** (Poisson's equation). *Poisson's equation* over some domain  $\mathcal{D}$  is

$$\nabla^2 \phi = f$$

where  $\phi(\mathbf{x})$  is an unknown scalar field and  $f(\mathbf{x})$  is some *source term*.

**Definition** (Laplace's equation). *Laplace's equation* over some domain  $\mathcal{D}$  is Poisson's equation with zero source term, that is

$$\nabla^2 \phi = 0$$

Of course, for such a differential equation, we need boundary conditions, which come in two main forms:

- *Dirichlet boundary conditions*. These simply give

$$\phi = g$$

for  $\mathbf{x} \in \partial \mathcal{D}$ ;

- *von Neumann boundary conditions*. These are that

$$\frac{\partial \phi}{\partial \mathbf{n}} = g$$

for  $\mathbf{x} \in \partial \mathcal{D}$ , where  $\frac{\partial \phi}{\partial \mathbf{n}}$  is the directional derivative  $\mathbf{n} \cdot \nabla \phi$  as discussed earlier.

These equations often arise in problems involving potentials. For example, in electrostatics, we use Maxwell's equations with everything independent of time. Then,  $\nabla \times \mathbf{E} = 0$ , so  $\mathbf{E}$  is irrotational and may be expressed (in a simply-connected domain) in terms of an electric potential  $\phi(\mathbf{x})$  as

$$\mathbf{E} = -\nabla \phi$$

We then obtain from Gauss's law

**Proposition 7.1**

$$\nabla^2 \phi = -\frac{\rho}{\varepsilon_0}$$

which is an example of Poisson's equation.

**Example**

Let us solve Laplace's equation in the domain  $0 \leq x, y \leq 1$  with Dirichlet boundary conditions

$$\begin{aligned} \phi(x, 0) &= 0 & \phi(0, y) &= 0 \\ \phi(x, 1) &= 0 & \phi(1, y) &= \sin \pi y \end{aligned}$$

We will guess a solution of the form  $\phi(x, y) = p(x)q(y)$ . From  $\phi(1, y)$  we see that  $q(y) = A \sin \pi y$  where  $A = \frac{1}{p(1)}$ . Hence

$$0 = \nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = p''(x)q(y) + p(x)q''(y) = A(p'' - \pi^2 p) \sin \pi y$$

so  $p(x) = B \sinh \pi x + C \cosh \pi x$ . We can match the boundary conditions by taking  $B = 1, C = 0$ , so a solution is

$$\phi(x, y) = \frac{\sinh \pi x \sin \pi y}{\sinh \pi}$$

We will show in 7.2 that solutions are unique, so this is *the* solution.

**Example**

Let us solve Poisson's equation  $\nabla^2 \phi = r$  in the domain  $r < 1$  in  $\mathbb{R}^3$  with Dirichlet boundary condition  $\phi = 0$  on  $r = 1$ . Because of the spherical symmetry, we will seek a solution  $\phi(r)$  dependent on  $r$  only. Recall

$$\nabla^2 \phi = \phi''(r) + \frac{2\phi'(r)}{r} = \frac{1}{r^2} \frac{d}{dr} (r^2 \phi'(r)) = \frac{1}{r} \frac{d^2}{dr^2} (r\phi(r))$$

I feel like using the final expression, so we have

$$\begin{aligned} \frac{1}{r}(r\phi)'' &= r \\ \implies r\phi &= \frac{1}{12}r^4 + Ar + B \\ \implies \phi &= \frac{1}{12}r^3 + A + \frac{B}{r} \end{aligned}$$

For  $\phi$  to be defined at  $r = 0$ , we must have  $B = 0$ . On the boundary  $r = 1$ ,  $\phi = 0$ , so  $A = -\frac{1}{12}$ . Hence  $\phi = \frac{1}{12}(r^3 - 1)$  is the solution.

### Example

Let us solve Laplace's equation in  $\mathbb{R}^3$  with  $1 < r < 2$  with mixed boundary conditions:  $\phi = 0$  on  $r = 1$  and  $\frac{\partial \phi}{\partial \mathbf{n}} = 1$  on  $r = 2$ . Again, let us try  $\phi = \phi(r)$ , so

$$\frac{1}{r}(r\phi)'' = 0 \implies \phi = A + \frac{B}{r}$$

On  $r = 1$ ,  $A + B = 0$ . On  $r = 2$ ,  $\mathbf{n} = \mathbf{e}_r$ , so  $\frac{\partial \phi}{\partial \mathbf{n}} = \mathbf{e}_r \cdot \nabla \phi = \frac{\partial \phi}{\partial r} \Big|_{r=2}$ , so  $-\frac{1}{4}B = 1$ . So the solution is  $\phi(r) = 4\left(1 - \frac{1}{r}\right)$ .

### §7.1.1 Superposition of solutions

The Laplacian operator is linear. So we can solve  $\nabla^2 \phi = f_1 + f_2$  using the superposition  $\phi = \phi_1 + \phi_2$ , where  $\phi_1$  and  $\phi_2$  solve  $\nabla^2 \phi_1 = f_1$  and  $\nabla^2 \phi_2 = f_2$  respectively.

One special case of this is that, to solve Poisson's equation with Dirichlet boundary conditions,

$$\nabla^2 \phi = f \qquad \phi = g \text{ on } \partial \mathcal{D}$$

it can sometimes be helpful to split the problem into two using

$$\begin{aligned} \nabla^2 \phi_1 &= f \text{ in } \mathcal{D} & \nabla^2 \phi_2 &= 0 \text{ in } \mathcal{D} \\ \phi_1 &= 0 \text{ on } \partial \mathcal{D} & \phi_2 &= g \text{ on } \partial \mathcal{D} \end{aligned}$$

### Lecture $n + 1$

**Remark** (Consistency for von Neumann boundary conditions). For the problem

$$\nabla^2 \phi = f \text{ in } \mathcal{D}, \qquad \frac{\partial \phi}{\partial \mathbf{n}} = g \text{ on } \partial \mathcal{D}$$

we have

$$\int_{\mathcal{D}} f \, dV = \int_{\mathcal{D}} \nabla \cdot (\nabla \phi) \, dV = \int_{\partial \mathcal{D}} \nabla \phi \cdot d\mathbf{S} = \int_{\partial \mathcal{D}} g \, dS$$

by the divergence theorem. Hence, if it is not true that  $\int_{\mathcal{D}} f \, dV = \int_{\partial \mathcal{D}} g \, dS$ , there is no solution.

### §7.2 Uniqueness of solutions

#### Proposition

The solution to Poisson's equation with Dirichlet boundary conditions is unique.

*Proof.* Suppose that  $\phi_1$  and  $\phi_2$  are both solutions to

$$\nabla^2 \phi = f \text{ in } \mathcal{D}, \qquad \phi = g \text{ on } \partial \mathcal{D}$$

Let  $\psi = \phi_1 - \phi_2$ . Then

$$\nabla^2 \psi = 0 \text{ in } \mathcal{D}, \quad \psi = 0 \text{ on } \partial\mathcal{D}$$

We apply the divergence theorem 6.3 to  $\mathbf{F} = \psi \nabla \psi$  to obtain

$$\begin{aligned} 0 &= \int_{\partial\mathcal{D}} \psi \nabla \psi \cdot d\mathbf{S} = \int_{\mathcal{D}} \nabla \cdot (\psi \nabla \psi) dV \\ &= \int_{\mathcal{D}} (\nabla \psi \cdot \nabla \psi + \underbrace{\psi \nabla^2 \psi}_0) dV \\ &= \int_{\mathcal{D}} |\nabla \psi|^2 dV \end{aligned} \quad (*)$$

Since  $|\nabla \psi|^2 \geq 0$ , we must have  $\nabla \psi = \mathbf{0}$  everywhere, so  $\psi$  must be a constant. Furthermore, it vanishes on  $\partial\mathcal{D}$ , so  $\psi = 0$  and  $\phi_1 = \phi_2$  as required. ■

### Proposition

The solution to Poisson's equation with von Neumann boundary conditions is unique up to an arbitrary additive constant.

*Proof.* We proceed exactly as above, until equation (\*). This equation still holds, since

$$\nabla \psi \cdot d\mathbf{S} = \frac{\partial \psi}{\partial \mathbf{n}} dS = 0$$

on  $\partial\mathcal{D}$ . Again, we obtain  $\nabla \psi = \mathbf{0}$  everywhere, so  $\psi$  is a constant  $C$ , and hence  $\phi_1 = \phi_2 + C$  as required. ■

**Remark** (Non-examinable). The proofs above require that  $\partial\mathcal{D}$  be closed so that we can use the divergence theorem, which means that  $\mathcal{D}$  must be finite. We sometimes want to use uniqueness in an *infinite* domain by imposing a boundary condition at infinity. Normally we can justify this by taking a limit of finite surfaces.

## §7.3 Gauss's flux method

Gauss's flux method is a nice way to solve Poisson's equation  $\nabla^2 \phi = f$  in radially symmetric situations.

### Theorem 7.2 (Gauss's flux method)

Consider the Poisson's equation

$$\nabla^2 \phi = f$$

where  $f = f(r) = f(|\mathbf{x}|)$ . Then

$$\phi = \int \left( \frac{1}{r^2} \int_0^r r'^2 f(r') dr' \right) dr$$

*Proof.* Put  $\phi = \phi(r)$ , so that  $\nabla\phi = \phi'(r)\mathbf{e}_r$ .

Consider a sphere of radius  $R$ , and let  $F_R$  be the total source strength within the sphere,

$$F_R = \int_{r \leq R} f(r) dV = \int_{r=0}^R \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} f(r) r^2 \sin \theta d\phi d\theta dr = 4\pi \int_0^R r^2 f(r) dr$$

Disturbingly, by the way, we are using  $\phi$  twice to mean completely different things. Anyways, we also have,

$$\begin{aligned} F_R &= \int_{r \leq R} \nabla^2 \phi dV \\ &= \int_{r=R} \nabla \phi \cdot d\mathbf{S} && \text{(divergence thm)} \\ &= \int_{r=R} \phi'(r) \mathbf{e}_r \cdot \mathbf{e}_r dS \\ &= \phi'(R) \int_{r=R} dS \\ &= 4\pi R^2 \phi'(R) \end{aligned}$$

Hence, we obtain

$$\phi'(R) = \frac{1}{R^2} \int_0^R r^2 f(r) dr$$

which, after integrating, gives the result. ■

**Remark.** If we impose the boundary condition  $\phi(r) \rightarrow 0$  as  $r \rightarrow \infty$ , then we can put limits on the indefinite integral and obtain

$$\phi(r) = - \int_r^\infty \left( \frac{1}{R^2} \int_0^R r'^2 f(r') dr' \right) dR$$

**Remark.** The final formula is not supposed to be memorised, and one will receive no marks for quoting it. Instead, it is necessary to derive it every time. Furthermore, we may often be asked to merely find  $\nabla\phi$ , in which case the final integration is unnecessary.

An analogous statement to Gauss's flux method holds for cylindrically symmetric  $f$ 's as well, considering the total source within a cylinder of radius  $R$  and height 1 will give a very similar result. We obtain

$$\phi'(R) = \frac{1}{R} \int_0^R \rho f(\rho) d\rho$$

**Example**

Let us find the electric potential produced by the charge density

$$\rho(r) = \begin{cases} \rho_0 & r \leq a \\ 0 & r > a \end{cases}$$

for some constant  $\rho_0$ . Recalling [proposition 7.1](#), we have  $\nabla^2\phi = -\rho/\varepsilon_0$ , so

$$f(r) = \begin{cases} -\rho_0/\varepsilon_0 & r \leq a \\ 0 & r > a \end{cases}$$

so

$$F_R = \int_{r \leq R} f(r) dV = \begin{cases} -\frac{\rho_0}{\varepsilon_0} \cdot \frac{4}{3}\pi R^3 & R \leq a \\ -\frac{\rho_0}{\varepsilon_0} \cdot \frac{4}{3}\pi a^3 & R > a \end{cases}$$

Thus

$$\phi'(R) = \frac{1}{4\pi R^2} F_R = \begin{cases} -\frac{\rho_0}{3\varepsilon_0} R & R \leq a \\ -\frac{\rho_0 a^3}{3\varepsilon_0 R^2} & R > a \end{cases}$$

which, remember, we would have to re-derive in the exam. Now

$$\phi(r) = \begin{cases} -\frac{\rho_0}{6\varepsilon_0} r^2 + c_1 & r \leq a \\ \frac{\rho_0 a^3}{3\varepsilon_0 r} + c_2 & r > a \end{cases}$$

Since  $\phi$  is continuous at  $a$  and  $\phi \rightarrow 0$  as  $r \rightarrow \infty$  by convention, we have  $c_2 = 0$ ,  $c_1 = \rho_0 a^2/2\varepsilon_0$ . We obtain

$$\phi(r) = \begin{cases} \frac{\rho_0}{6\varepsilon_0} (3a^2 - r^2) & r \leq a \\ \frac{\rho_0 a^3}{3\varepsilon_0 r} & r > a \end{cases}$$

We can also calculate  $\mathbf{E} = -\nabla\phi$  by just multiplying our expression for  $\phi'(R)$  by  $-\mathbf{e}_r$ .

### Example

Consider a sphere of radius  $a$  centred at the origin, with uniform charge density except for a hollow void of radius  $\frac{1}{2}a$  centred at a distance  $\frac{1}{3}a$  from the origin, with total charge  $Q$ . What is the potential outside the sphere?

Sadly, this situation is not spherically symmetric, so one cannot automatically use Gauss's method.

First, let us find the uniform charge density, which is

$$\rho_0 = \frac{Q}{\frac{4}{3}\pi a^3 - \frac{4}{3}\pi \left(\frac{1}{2}a\right)^3} = \frac{6Q}{7\pi a^3}$$

Our method for this will be to solve the problem for the entire sphere with charge density  $\rho_0$ , then add the smaller sphere in with charge density  $-\rho_0$ , centred at  $\mathbf{x}_0 = (\frac{1}{3}a, 0, 0)$ . Quoting the formula from the previous example, we obtain

$$\phi = \frac{\rho_0 a^3}{3\epsilon_0 |\mathbf{x}|} - \frac{\rho_0 \left(\frac{1}{2}a\right)^3}{3\epsilon_0 |\mathbf{x} - \mathbf{x}_0|} = \frac{2Q}{7\pi\epsilon_0} \left( \frac{1}{|\mathbf{x}|} - \frac{1}{8|\mathbf{x} - \mathbf{x}_0|} \right)$$

## §7.4 Point sources

Consider Poisson's equation with a spherically symmetric source term which is zero everywhere except in a region  $r \leq \varepsilon$ , with the total source strength being  $F = \int_{r \leq \varepsilon} f dV$ . Using Gauss's method, we obtain

$$\phi(r) = -\frac{F}{4\pi r}$$

for any  $r \geq \varepsilon$ . Now, we are free to shrink the source to a point, keeping  $F$  fixed, to obtain  $\phi(r)$  as the field produced by a point source everywhere.

### Lecture $n + 1$ – 06/03/26

In the case of electric charge, where  $f = -\rho/\epsilon_0$ , we have  $F = -Q/\epsilon_0$ , which gives Coulomb's law:

$$\phi = \frac{Q}{4\pi\epsilon_0 r}, \quad \mathbf{E} = -\nabla\phi = \frac{Q}{4\pi\epsilon_0 r^2} \mathbf{e}_r$$

The force on any test charge  $q$  placed in this field is  $q\mathbf{E}$ . In other words, Coulomb's law is a consequence of Maxwell's equations.

The equivalent solution for a *line source* along the  $z$ -axis of strength  $F$  per unit length is

$$\phi = \frac{F}{2\pi} \log \rho$$

and is also the solution for a point source in 2D.

**Remark** (Non-examinable). A point source of strength  $F$  at the origin can be considered as a source field  $\mathbf{f}(\mathbf{x}) = F\delta(\mathbf{x})$ , where  $\delta(\mathbf{x}) = \delta(x)\delta(y)\delta(z)$ . Hence we have

$$\nabla^2 \left( -\frac{F}{4\pi r} \right) = F\delta(\mathbf{x})$$

This is an example of the mythical *Green's functions*.

## §7.5 Gravitation

Consider a point mass  $M$  at the origin. A test mass  $m$  in its gravitational field experiences a force

$$-\frac{GMm}{r^2}\mathbf{e}_r$$

so the potential field created by  $M$  is  $\phi = -GM/r$  *per unit mass*. This is exactly the same situation as for electric point charges, up to a constant. Hence, it is equivalent to the Poisson's equation

$$\nabla^2\phi = 4\pi G\rho$$

where  $\rho$  is the mass density.

### Example

Let us take a hollow sphere with mass density

$$\rho = \begin{cases} \rho_0 & a \leq r \leq b \\ 0 & \text{o/w} \end{cases}$$

We would like to show that the gravitational field vanishes inside the cavity.

Since the system is spherically symmetric, we can use Gauss's method with

$$F_R = \int_{r \leq R} 4\pi G\rho dV = 0$$

if  $R < a$ . Hence  $\phi'(R) = \frac{1}{4\pi R^2}F_R = 0$  for  $R < a$  (except perhaps at  $R = 0$ , but, since  $\nabla\phi$  must be continuous, we can avoid worrying about that). It immediately follows that  $-\nabla\phi = \phi'(r)\mathbf{e}_r = 0$  inside the cavity.

**Remark** (Non-examinable). Another way of dealing with this problem would be to superpose spheres, as we did for charges earlier.

## §7.6 The integral solution of Poisson's equations

We have seen that the solution of Poisson's equation for a point source  $F$  is  $-\frac{F}{4\pi r}$ .

Now, consider a source field  $f(\mathbf{x})$  confined to some finite region  $V \subset \mathbb{R}^3$ . Let us divide  $V$  up into infinitesimal volumes  $dV$ , each with position  $\mathbf{y}$ . Each such volume acts as a point source of strength  $f(\mathbf{y}) dV$  with corresponding potential  $-\frac{f(\mathbf{y})}{4\pi|\mathbf{x}-\mathbf{y}|} dV$ . Hence, by superposition, we obtain

$$\phi(\mathbf{x}) = \int_V \frac{f(\mathbf{y})}{4\pi|\mathbf{x}-\mathbf{y}|} d^3\mathbf{y}$$

## §7.7 Harmonic functions and the maximum principle

**Definition** (Harmonic functions). A *harmonic function* is one that obeys Laplace's equation in some domain.

### Proposition

Given a function  $\phi(\mathbf{x})$  that is harmonic in  $V \subseteq \mathbb{R}^3$ , let  $B_r(\mathbf{a})$  be a sphere of radius  $r$  centred at  $\mathbf{a} \in V$ , and assume that  $B_r(\mathbf{a}) \subseteq V$ . Let

$$\bar{\phi}_r(\mathbf{a}) = \frac{1}{4\pi r^2} \int_{\partial B_r(\mathbf{a})} \phi(\mathbf{x}) dS$$

be the average value of  $\phi$  over the surface of the sphere. Then

$$\bar{\phi}_r(\mathbf{a}) = \phi(\mathbf{a})$$

*Proof.* We will use spherical polar coordinates centred at  $\mathbf{a}$ , i.e.

$$\mathbf{x} = \mathbf{a} + \begin{pmatrix} r \sin \theta \cos \phi \\ r \sin \theta \sin \phi \\ r \cos \theta \end{pmatrix}$$

so that  $\mathbf{e}_r$  is a unit vector pointing radially away from  $\mathbf{a}$ . Now

$$\begin{aligned} \bar{\phi}_r(\mathbf{a}) &= \frac{1}{4\pi r^2} \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \phi(\mathbf{x}) r^2 \sin \theta d\theta d\phi \\ &= \frac{1}{4\pi} \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \phi(\mathbf{x}) \sin \theta d\theta d\phi && (r \text{ const.}) \\ \implies \frac{d}{dr} \bar{\phi}_r(\mathbf{a}) &= \frac{1}{4\pi} \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \frac{\partial \phi}{\partial r} \sin \theta d\theta d\phi \\ &= \frac{1}{4\pi r^2} \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \nabla \phi \cdot \mathbf{e}_r r^2 \sin \theta d\theta d\phi \\ &= \frac{1}{4\pi r^2} \int_{\partial B_r(\mathbf{a})} \nabla \phi \cdot d\mathbf{S} \\ &= \frac{1}{4\pi r^2} \int_{B_r(\mathbf{a})} \nabla^2 \phi dV && (\text{divergence thm}) \\ &= 0 \end{aligned}$$

Hence  $\bar{\phi}_r(\mathbf{a})$  is constant – but clearly, as  $r \rightarrow 0$ , it tends to  $\phi(\mathbf{a})$ , since  $\phi$  is continuous. The result follows.  $\blacksquare$

**Remark.** By integrating  $4\pi r^2 \bar{\phi}_r(\mathbf{a})$  from 0 to  $r$ , we can further show that

$$\phi(\mathbf{a}) = \frac{1}{\frac{4}{3}\pi r^3} \int_{B_r(\mathbf{a})} \phi(\mathbf{x}) dV$$

### Proposition (The maximum principle)

Let  $\phi$  be harmonic in  $V \subset \mathbb{R}^3$ . Then its maximum value is achieved on  $\partial V$ .

*Proof.* Suppose that  $\phi$  achieves its maximum at some interior point  $\mathbf{a}$ . Then  $\phi(\mathbf{x}) \leq \phi(\mathbf{a})$  for all  $\mathbf{x} \in V$ . Now consider  $\partial B_r(\mathbf{a})$ . If there were a point  $\mathbf{x} \in \partial B_r(\mathbf{a})$  such that

$\phi(\mathbf{x}) < \phi(\mathbf{a})$ , then the average value  $\bar{\phi}_r(\mathbf{a})$  would be strictly less than  $\phi(\mathbf{a})$ , contradicting the proposition. Hence  $\phi(\mathbf{x}) = \phi(\mathbf{a})$  for all  $\mathbf{x} \in \partial B_r(\mathbf{a})$ .

Now we increase  $r$  until it is large enough for  $\partial B_r(\mathbf{a})$  to intersect the boundary  $\partial V$  at some point  $\mathbf{b}$ . Then  $\phi(\mathbf{b}) = \phi(\mathbf{a})$ , so  $\phi$  achieves its maximum over  $V$  at  $\mathbf{b} \in \partial V$ . ■

Of course, the same is true for the minimum.

**Remark.** This is in fact the *weak* maximum principle. The *strong* maximum principle states that  $\phi$  *only* achieves its maximum on  $\partial V$ , *not* at any interior point, unless  $\phi$  is in fact constant throughout  $V$ .

## §8 Cartesian tensors

In this chapter,  $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$  will refer to any general right-handed orthonormal basis of  $\mathbb{R}^3$ , not just the standard basis, as will  $\{\mathbf{e}'_1, \mathbf{e}'_2, \mathbf{e}'_3\}$ .

### §8.1 What is a vector?

A vector is *not* three numbers in a column – it has an existence of its own within a vector space. An equation like  $\mathbf{v} = (1, 2, 4)^\top$  merely refers to the components of  $\mathbf{v}$  with respect to one particular basis – if we change the basis, the components of  $\mathbf{v}$  will change, while  $\mathbf{v}$  itself will remain exactly the same. This is also called changing the *frame of reference*.

#### Proposition 8.1 (Transformation rule for vectors)

Let  $\mathbf{v}$  have components  $v_j$  with respect to a basis  $\{\mathbf{e}_i\}$  and  $v'_i$  with respect to a basis  $\{\mathbf{e}'_i\}$ . Then

$$v'_i = R_{ij}v_j$$

where  $R$  is a particular matrix.

**Definition 8.2.** For bases  $\{\mathbf{e}_j\}$  and  $\{\mathbf{e}'_i\}$ , the matrix  $R$  satisfies

$$R_{ij} = \mathbf{e}'_i \cdot \mathbf{e}_j$$

*Proof.* We have

$$\mathbf{v} = v_j \mathbf{e}_j = v'_k \mathbf{e}'_k$$

so, dotting both sides with  $\mathbf{e}'_i$ ,

$$v'_k \delta_{ik} = v_j \mathbf{e}'_i \cdot \mathbf{e}_j$$

from which the result follows. ■

Note that

$$R = \begin{pmatrix} \mathbf{e}'_1 \cdot \mathbf{e}_1 & \mathbf{e}'_1 \cdot \mathbf{e}_2 & \mathbf{e}'_1 \cdot \mathbf{e}_3 \\ \mathbf{e}'_2 \cdot \mathbf{e}_1 & \mathbf{e}'_2 \cdot \mathbf{e}_2 & \mathbf{e}'_2 \cdot \mathbf{e}_3 \\ \mathbf{e}'_3 \cdot \mathbf{e}_1 & \mathbf{e}'_3 \cdot \mathbf{e}_2 & \mathbf{e}'_3 \cdot \mathbf{e}_3 \end{pmatrix} = \begin{pmatrix} - & \mathbf{e}'_1{}^\top & - \\ - & \mathbf{e}'_2{}^\top & - \\ - & \mathbf{e}'_3{}^\top & - \end{pmatrix}$$

where we are expressing each of the basis vectors with their components in the original basis  $\{\mathbf{e}_i\}$ . Since the three rows of  $R$  are orthonormal vectors, then,  $R$  is an orthogonal matrix. If we really want to get into the weeds,

$$R^T R = \begin{pmatrix} -\mathbf{e}_1^T & - \\ -\mathbf{e}_2^T & - \\ -\mathbf{e}_3^T & - \end{pmatrix} \begin{pmatrix} | & | & | \\ \mathbf{e}'_1 & \mathbf{e}'_2 & \mathbf{e}'_3 \\ | & | & | \end{pmatrix} = \begin{pmatrix} \mathbf{e}'_1 \cdot \mathbf{e}'_1 & \mathbf{e}'_1 \cdot \mathbf{e}'_2 & \mathbf{e}'_1 \cdot \mathbf{e}'_3 \\ \mathbf{e}'_2 \cdot \mathbf{e}'_1 & \mathbf{e}'_2 \cdot \mathbf{e}'_2 & \mathbf{e}'_2 \cdot \mathbf{e}'_3 \\ \mathbf{e}'_3 \cdot \mathbf{e}'_1 & \mathbf{e}'_3 \cdot \mathbf{e}'_2 & \mathbf{e}'_3 \cdot \mathbf{e}'_3 \end{pmatrix} = I$$

Anyhow, this allows us to conclude that

$$R_{ij}R_{jk} = \delta_{ik} \quad R_{ki}R_{kj} = \delta_{ij}$$

Furthermore, expanding the determinant along the first row, we have

$$\det R = \det \begin{pmatrix} -\mathbf{e}_1^T & - \\ -\mathbf{e}_2^T & - \\ -\mathbf{e}_3^T & - \end{pmatrix} = \mathbf{e}'_1 \cdot (\mathbf{e}'_2 \times \mathbf{e}'_3) = 1$$

since the  $\{\mathbf{e}'_i\}$  are a right-handed basis – so  $R$  is in fact a rotation matrix. In particular, anything that does not obey [proposition 8.1](#) is not a vector!

We also have

### Proposition 8.3 (Inverse transformation rule for vectors)

Let  $\mathbf{v}$  have components  $v_i$  with respect to a basis  $\{\mathbf{e}_i\}$  and  $v'_j$  with respect to a basis  $\{\mathbf{e}'_i\}$ . Then

$$v_i = R_{ji}v'_j$$

where  $R$  is the same matrix.

*Proof.* This is pretty clear. If we want to get annoying about it,

$$\begin{aligned} v'_j &= R_{jk}v_k \\ \implies R_{ji}v'_j &= R_{ij}R_{jk}v_k \\ &= (R^T R)_{ik}v_k \\ &= \delta_{ik}v_k = v_i \end{aligned}$$

■

**Remark.** These transformation rules look like a rotation map between two different vectors  $\mathbf{v}$  and  $\mathbf{v}'$ . This is misleading, however, because  $v'_i$  and  $v_j$  are components of the *same* vector in two different frames.

## §8.2 What is a tensor?

**Definition 8.4.** A *tensor*  $T$  of rank or order  $n$  is an array of numbers  $T_{ijk\dots}$  (with respect to a given basis) with  $n$  indices which, under a change of basis, transforms according to

$$T'_{ijk\dots} = R_{ip}R_{jq}R_{kr}\cdots T_{pqr\dots}$$

where  $T'_{ijk\dots}$  are the components of  $T$  with respect to the new basis, and  $R$  is the relevant rotation matrix as defined in [definition 8.2](#). Note that rank must not be confused with dimension (we will work exclusively in 3D, so each of the indices take values 1, 2, 3).

**Remark.**  $T_{ijk\dots}$  can be thought of as a 3 by 3 by ... by 3 matrix ( $n$  times).

### Lemma 8.5

The inverse of the defining property of tensors is

$$T_{ijk\dots} = R_{pi}R_{qj}R_{rk}\cdots T'_{pqr}$$

**Example** • A tensor of rank 1 satisfies, according to [definition 8.4](#),  $T'_i = R_{ip}T_p$ , so tensors of rank 1 are vectors.

- A tensor of rank 0 satisfies  $T' = T$ , so it is a real number invariant under change of basis. In other words, it is a scalar.

### Proposition 8.6

A tensor of rank 2 satisfies

$$T'_{ij} = R_{ip}R_{jq}T_{pq}$$

### Proposition 8.7

A tensor of rank 2 is just a matrix. It satisfies

$$T' = RTR^\top$$

**Remark.** One can define tensors with respect to other orthogonal curvilinear coordinate systems, such as the spacetime of general relativity. In this context we may encounter *covariant* vectors  $v_i$  and *contravariant* vectors  $v^i$ , a distinction which is non-existent in Cartesians.

## §8.3 Examples of tensors

- Temperature,  $\Theta$ , is a scalar – it is invariant under a change of basis.
- Position, velocity and acceleration are tensors of rank 1.
- The Kronecker delta is a tensor of rank 2 which, by definition, is the same in every basis,  $\delta'_{ij} = \delta_{ij}$ . Indeed,

$$R_{ip}R_{jq}\delta_{pq} = R_{ip}R_{jp} = (RR^\top)_{ij} = \delta_{ij} = \delta'_{ij}$$

or, alternatively, using [proposition 8.7](#),

$$RTR^\top = RIR^\top = RR^\top = I = T'$$

where  $T$  is the matrix of the tensor corresponding to  $\delta_{ij}$ , namely the identity.

- The Levi-Civita symbol  $\varepsilon_{ijk}$  is a tensor of rank 3, again defined to be the same for every basis. To verify:

$$R_{ip}R_{jq}R_{kr}\varepsilon_{pqr} = \varepsilon_{ijk} \det R = \varepsilon_{ijk} = \varepsilon'_{ijk}$$

since  $R$  is a rotation matrix. Hence  $\varepsilon_{ijk}$  is a tensor.

- Conductivity is a tensor of rank 2. The basic conductivity law is  $\mathbf{J} = \sigma \mathbf{E}$ , where  $\mathbf{E}$  is the applied electric field and  $\mathbf{J}$  is the resulting electric current. In an isotropic medium,  $\sigma$  is just a scalar. In anisotropic media, it makes sense to represent  $\sigma$  as a matrix, and it should clearly obey the tensor transformation law. In particular,  $\mathbf{J}$  obeys the tensor transformation law for vectors if and only if  $\sigma$  obeys the tensor transformation law for matrices by the quotient theorem, which we will see later.

What about things that aren't tensors?

- An object given by  $\mathbf{w} = (0, 0, 0)$  in one frame and  $(1, 0, 0)$  is not a vector.
- An object given by  $\delta_{ij}$  in one frame and  $-\delta_{ij}$  in another frame is not a tensor.
- Let  $\alpha$  be the  $x$ -component of a non-zero vector. Then  $\alpha$  is not a scalar, because it will be different in different frames.

## §8.4 Sums and products of tensors

Results pertaining to general rank tensors are messy. Instead, we will show results for particular cases, which will easily generalise.

### §8.4.1 Linear combinations

#### Proposition

A linear combination of two tensors of the same rank is another tensor of the same rank.

*Proof.* Proof (rank 2) Let  $T_{ij} = \alpha A_{ij} + \beta B_{ij}$ . Then

$$\begin{aligned} T'_{ij} &= \alpha' A'_{ij} + \beta' B'_{ij} \\ &= \alpha' R_{ip}R_{jq}A_{pq} + \beta' R_{ip}R_{jq}B_{pq} \\ &= R_{ip}R_{jq}T_{pq} \end{aligned}$$

as required. ■

**Remark.** We need  $\alpha$  and  $\beta$  to be scalars, not just numbers.

### §8.4.2 Outer products

**Definition** (Outer products). The *outer product* of two tensors  $A_{ijk\dots}$  and  $B_{lmn\dots}$  of ranks  $m$  and  $n$  respectively is given by

$$T_{ijk\dots lmn\dots} = A_{ijk\dots} B_{lmn\dots}$$

and is a tensor of rank  $m + n$ . This can be denoted  $T = A \otimes B$ .

This is itself a tensor. For example, suppose  $A_{ij}$  and  $B_{ij}$  are second-rank tensors, and consider  $T_{ijkl} = A_{ij}B_{kl}$ . We have

$$\begin{aligned} T'_{ijkl} &= A'_{ij}B'_{kl} \\ &= R_{ip}R_{jq}A_{pq}R_{kr}R_{ls}B_{rs} \\ &= R_{ip}R_{jq}R_{kr}R_{ls}T_{pqrs} \end{aligned}$$

as required. Similarly, if  $\mathbf{v}$  and  $\mathbf{w}$  are vectors, then  $v_iw_j$  is a second-rank tensor.

### §8.4.3 Contraction

**Definition (Contraction).** A *contraction* of a rank  $n$  tensor occurs when we make two indices equal and sum, producing a tensor of rank  $n - 2$ .

This is also still a tensor. For example, if  $T_{ij}$  is a tensor of rank 2, then  $\alpha = T_{ii}$  is a scalar, because

$$\alpha' = T'_{ii} = R_{ip}R_{iq}T_{pq} = (R^T R)_{pq}T_{pq} = \delta_{pq}T_{pq} = T_{pp} = \alpha$$

### §8.4.4 Inner products

**Definition (Inner products).** The *inner product* of two tensors  $A_{ijk\dots}$  and  $B_{lmn\dots}$  of ranks  $m$  and  $n$  respectively is given by multiplying them, setting one index in each equal, and summing, resulting in a tensor of rank  $m + n - 2$ . This is the same as taking the outer product and then contracting, so it is also known as *contracting  $A$  and  $B$  together*.

For example, if  $v_i$  and  $w_i$  are vectors,  $v_iw_i$  is a scalar; if  $A_{ij}$  is a rank 2 tensor and  $v_i$  is a vector, then  $A_{ij}v_j$  is a vector, as is  $A_{ijk}v_i$ ; if  $A_{ijk}$  and  $B_{ij}$  are rank 3 and 2 tensors, then  $A_{ijk}B_{lj}$  is a third-rank tensor.

That these are actually tensors follows from the outer product followed by contraction.

**Example** • For two vectors  $\mathbf{u}$  and  $\mathbf{v}$ ,  $\mathbf{u} \cdot \mathbf{v}$  is a scalar, so defining the angle between them as  $\cos \theta = (\mathbf{u} \cdot \mathbf{v}) / \sqrt{(\mathbf{u} \cdot \mathbf{u})(\mathbf{v} \cdot \mathbf{v})}$  makes it a scalar.

- If  $\mathbf{a}$  and  $\mathbf{b}$  are vectors, then  $\mathbf{a} \times \mathbf{b}$  is a vector, since it is a double contraction of three tensors given by  $\varepsilon_{ijk}a_jb_k$ .

## §8.5 The quotient theorem

### Theorem (Quotient theorem)

Let  $T_{ijk\dots}$  be an object of some rank. If the product (inner or outer) of  $T$  with an *arbitrary* tensor  $A$  of a given rank is a tensor, then  $T$  itself is a tensor.

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*Proof.* We will show the result for  $T$  being an object of rank two, and taking an inner product with a tensor of rank one, namely a vector  $\mathbf{v}$ . Let  $w_i = T_{ij}v_j$ . By assumption,  $\mathbf{w}$  is a vector, so

$$T'_{ij}v'_j = w'_i = R_{ip}w_p$$

$$\begin{aligned}
&= R_{ip} T_{pq} v_q \\
&= R_{ip} T_{pq} R_{jq} v'_j \\
\implies 0 &= v'_j (T'_{ij} - R_{ip} R_{jq} T_{pq})
\end{aligned}$$

Since this is true for arbitrary  $v'_j$ , we can cancel, so  $T'_{ij} = R_{ip} R_{jq} T_{pq}$  as required.  $\blacksquare$

Let's do another one for practice. Suppose  $T$  is an object of rank two, and take an inner product with a tensor of rank one. We have

$$\begin{aligned}
T'_{ki} A'_{jk} &= W'_{ij} = R_{ip} R_{jq} W_{pq} \\
&= R_{ip} R_{jq} T_{rp} A_{qr} \\
&= R_{ip} R_{jq} T_{rp} R_{sq} R_{kr} A'_{sk} \\
&= R_{ip} T_{rp} R_{kr} \delta_{js} A'_{sk} \\
\implies 0 &= (T'_{ki} - R_{kr} R_{ip} T_{rp}) A'_{jk} = 0
\end{aligned}$$

so, once again, we can cancel, and obtain the transformation law for  $T$ .

**Remark** (Non-examinable). Let us think a bit more about what tensors really are. Since we understand they shouldn't be thought of merely as an array of numbers, let's instead view them as multilinear maps. Given a tensor  $T_{ijk\dots}$  of rank  $n$ , consider the map

$$\begin{aligned}
T &: (\mathbb{R}^3)^n \rightarrow \mathbb{R} \\
(\mathbf{a}, \mathbf{b}, \mathbf{c}, \dots) &\mapsto T_{ijk\dots} a_i b_j c_k \dots
\end{aligned}$$

This is multilinear and an inner product, so outputs a scalar – in other words, it is well-defined. For each tensor, such a map exists. Conversely, given such a map, multilinearity shows that it must be of the form above for some coefficients  $T_{ijk\dots}$ , which can be computed in any basis – for example,  $T_{111\dots} = T(\mathbf{e}_1, \mathbf{e}_1, \mathbf{e}_1, \dots)$ . The quotient theorem then says that  $T_{ijk\dots}$  is in fact a tensor. Hence, there is a complete equivalence between tensors and multilinear maps.

In particular, this can help us understand why the transformation rule for tensors of rank 2 looks just like the change of basis formula for linear maps –  $A' = P^\top A P$ , where  $P$  is the orthogonal change of basis matrix. The only difference is that  $P = R^\top = R^{-1}$ .

But why is  $P$  given by  $R^{-1}$  instead of  $R$ ? Consider a vector  $\mathbf{v}$  in 2D. Suppose that  $\{\mathbf{e}'_1, \mathbf{e}'_2\}$  are the original basis vectors  $\{\mathbf{e}_1, \mathbf{e}_2\}$  rotated through  $\theta$ , so that  $P$  is a rotation matrix with angle  $\theta$ . Then (a) finding the components of  $\mathbf{v}$  w.r.t.  $\{\mathbf{e}'_1, \mathbf{e}'_2\}$  is the same as (b) rotating  $\mathbf{v}$  by  $-\theta$  and finding its components w.r.t.  $\{\mathbf{e}_1, \mathbf{e}_2\}$ . Then, the tensor transformation rule for vectors represents (b), with  $R$  being a rotation by  $-\theta$ , while  $P$  represents the change of basis needed for (a).

## §8.6 Symmetries of tensors

If  $T_{ij}$  is a tensor, so is  $T_{ji}$ . Let  $U_{ij} = T_{ji}$ , so

$$U'_{ij} = T'_{ji} = R_{jp} R_{iq} T_{pq} = R_{iq} R_{jp} U_{qp} = R_{ip} R_{jq} U_{pq}$$

as required.

**Remark.** When trying to prove such things, it's best to take this approach of coming up with a new name for the new tensor.

**Definition** (Symmetric tensors). A tensor  $T_{ijk\dots}$  is *symmetric* in  $i$  and  $j$  if  $T_{ijk\dots} = T_{jik\dots}$ . The two indices need not be the first two, or adjacent.

For example, symmetric tensors correspond to symmetric matrices, since  $T_{ij} = T_{ji}$ .

It makes sense to say that a tensor is symmetric because this property is invariant under a change of basis. Suppose, for example, that  $T_{ij}$  is symmetric – we wish to show that  $T'_{ij}$  is symmetric as well. Observe

$$\begin{aligned} T'_{ij} &= R_{ip}R_{jq}T_{pq} = R_{ip}R_{jq}T_{qp} \\ &= R_{jq}R_{ip}T_{qp} \\ &= T'_{ji} \end{aligned}$$

as required.

**Definition (Antisymmetric tensors).** A tensor  $T_{ijk\dots}$  is *symmetric* in  $i$  and  $j$  if  $T_{ijk\dots} = -T_{jik\dots}$ . As before, any two indices can be chosen.

One can check for oneself that antisymmetry is also preserved under the tensor transformation law.

### Example

The Kronecker delta is symmetric of rank 2 – the Levi-Civita epsilon is antisymmetric in any two indices, so *totally antisymmetric*

### Proposition

The sum over the product of two tensors with a shared pair of indices which is symmetric in one and antisymmetric in the other is zero.

*Proof.* We show the result for rank 2. Observe

$$\begin{aligned} S_{ij}A_{ij} &= -S_{ji}A_{ji} \\ &= -S_{ij}A_{ij} \end{aligned}$$

by symmetry/antisymmetry and swapping dummy suffices. The result follows. ■

## §8.6.1 Decomposition of second-rank tensors

### Proposition 8.8 (Decomposition of second-rank tensors)

Any tensor  $T_{ij}$  of rank 2 can be written uniquely in the form

$$T_{ij} = S_{ij} + A_{ij}$$

where  $S_{ij}$  is symmetric and  $A_{ij}$  is antisymmetric.

### Proposition 8.9

In particular,

$$S_{ij} = \frac{1}{2} (T_{ij} + T_{ji}) \quad A_{ij} = \frac{1}{2} (T_{ij} - T_{ji})$$

*Proof.* Clearly  $S_{ij}$  and  $A_{ij}$  as above satisfy the requirements. Furthermore, supposing  $T_{ij} = S_{ij} + A_{ij}$ ,

$$T_{ij} + T_{ji} = S_{ij} + A_{ij} + S_{ji} + A_{ji} = 2S_{ij}$$

and  $T_{ij} - T_{ji} = 2A_{ij}$  similarly, so these are the only choices. ■

**Remark.**  $S$  and  $A$  are also tensors.

### Proposition

An antisymmetric second-rank tensor  $A_{ij}$  can be written in the form

$$\varepsilon_{ijk}\omega_k$$

for some vector  $\omega$ .

*Proof.* Let  $\omega_i = \frac{1}{2}\varepsilon_{ijk}A_{jk}$ . Then

$$\begin{aligned}\varepsilon_{ijk}\omega_k &= \frac{1}{2}\varepsilon_{ijk}\varepsilon_{klm}A_{lm} \\ &= \frac{1}{2}(\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl})A_{lm} \\ &= \frac{1}{2}(A_{ij} - A_{ji}) \\ &= A_{ij}\end{aligned}$$

where the last line is by antisymmetry of  $A$ . ■

Hence it follows that any second-rank tensor  $T$  can be represented as

$$T_{ij} = S_{ij} + \varepsilon_{ijk}\omega_k$$

for some symmetric tensor  $S$  and vector  $\omega$ .

**Remark.** Let

$$A = \begin{pmatrix} 0 & A_{12} & A_{13} \\ -A_{12} & 0 & A_{23} \\ -A_{13} & -A_{23} & 0 \end{pmatrix}$$

Then

$$\omega = \begin{pmatrix} A_{23} \\ -A_{13} \\ A_{12} \end{pmatrix}$$

Since second-rank antisymmetric tensors only have 3 degrees of freedom, it is perhaps not surprising that it can be expressed in terms of a vector.

**Example (hard)**

Suppose that  $C$  is an object of rank 2 such that  $C_{ij}S_{ij}$  is a scalar for any symmetric tensor  $S$  of rank 2. We claim that  $C$  may not be a tensor, but its symmetric part is.

Let  $T$  be an arbitrary second-rank tensor, and decompose it as  $T_{ij} = S_{ij} + A_{ij}$ ; also decompose  $C$  as  $C_{ij} = C_{ij}^{(S)} + C_{ij}^{(A)}$ , where the parts are symmetric and antisymmetric respectively. Then

$$\begin{aligned}
 C_{ij}^{(S)}T_{ij} &= C_{ij}^{(S)}(S_{ij} + A_{ij}) \\
 &= C_{ij}^{(S)}S_{ij} + \underbrace{C_{ij}^{(S)}A_{ij}}_0 \\
 &= (C_{ij} - C_{ij}^{(A)})S_{ij} \\
 &= C_{ij}S_{ij} - \underbrace{C_{ij}^{(A)}S_{ij}}_0 \\
 &= C_{ij}S_{ij}
 \end{aligned}$$

with repeated application of the fact that the product of symmetric and antisymmetric tensors is zero. We have obtained a scalar, i.e. a rank zero tensor. It follows that  $C_{ij}^{(S)}$  is a tensor by the quotient theorem.

However, we would like to exhibit a  $C$  which is not a tensor, and in particular we want  $C_{ij}^{(A)}$  not to be a tensor. Let  $C_{ij} = C_{ij}^A = \varepsilon_{1ij}$  in one frame and 0 in every other frame. But then

$$\begin{aligned}
 C_{ij}S_{ij} &= \begin{cases} \varepsilon_{1ij}S_{ij} & \text{first frame} \\ 0 & \text{other frames} \end{cases} \\
 &= 0
 \end{aligned}$$

**Remark.** In fact,  $\varepsilon_{1ij}$  is not a second-rank tensor, so we could even have made  $C_{ij} = \varepsilon_{1ij}$  in all frames.

**§8.7 Principal axes**

For any symmetric second-rank tensor  $S$ , we can choose a basis in which

$$(S_{ij}) = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}$$

which is of course just the same fact as that any real symmetric matrix can be diagonalised using an orthogonal change of basis matrix, namely a  $P$  such that  $P^\top SP = D$ . We let  $R = P^\top$  be our corresponding rotation matrix, which gives

$$S' = RSR^\top = P^\top SP = D$$

Note that, in general,  $P$  might have determinant  $\pm 1$ , but we can always re-order the columns of  $R$  to make it a rotation.

Of course, the  $\lambda_i$  are just the eigenvalues, but, for a tensor, they are known as *principal values* – and the *principal axes* are a set of orthonormal right-handed axes along each of the eigenvectors. Often, calculations are easier in the frame of the principal axes.

**Example**

Let us consider the *inertia tensor* for a rigid body  $V$ . Suppose the body has density  $\rho(\mathbf{x})$  and is rotating with angular velocity  $\boldsymbol{\omega}$ , so the mass in each infinitesimal volume,  $\rho(\mathbf{x}) dV$  has velocity  $\boldsymbol{\omega} \times \mathbf{x}$ . Hence, the total angular momentum is

$$\begin{aligned} \mathbf{L} &= \int_V \rho(\mathbf{x})(\mathbf{x} \times (\boldsymbol{\omega} \times \mathbf{x})) dV \\ &= \int_V \rho(\mathbf{x}) ((\mathbf{x} \cdot \mathbf{x})\boldsymbol{\omega} - (\boldsymbol{\omega} \cdot \mathbf{x})\mathbf{x}) dV \\ &= \int_V \rho(\mathbf{x}) (x_k x_k \omega_i - \omega_j x_j x_i) dV \\ &= I_{ij} \omega_i \end{aligned}$$

where

$$I_{ij} = \int_V \rho(\mathbf{x}) (x_k x_k \delta_{ij} - x_i x_j) dV$$

which is called the *inertia tensor*, a property of the body independent of  $\boldsymbol{\omega}$ . Since  $\boldsymbol{\omega}$  and  $\mathbf{L}$  are vectors, the relationship  $L_i = I_{ij} \omega_j$  shows that  $I_{ij}$  is indeed a tensor by the quotient theorem, and it is symmetric by definition (alternatively, we are taking products, multiplying by scalars, adding and integrating tensors, which must give a tensor).

This is also the inertia tensor about the origin – about a general point  $\mathbf{a}$  it is unsurprisingly given by

$$I_{ij}(\mathbf{a}) = \int_V \rho(\mathbf{x}) (|\mathbf{x} - \mathbf{a}|^2 \delta_{ij} - (x_i - a_i)(x_j - a_j)) dV$$

Clearly, the components  $I_{11}$ ,  $I_{22}$  and  $I_{33}$  of the inertia tensor are just the moments of inertia round the  $x$ ,  $y$  and  $z$  axes respectively.

**Example (hard)**

Let us find the inertia tensor of a cube of mass  $M$ , side length zero, with one corner at the origin and one at  $(a, a, a)$ . We have

$$\begin{aligned} I_{12} &= \int_V \frac{M}{a^3} (\underbrace{x_k x_k \delta_{12}}_0 - x_1 x_2) dV \\ &= -\frac{M}{a^3} \int_0^a \int_0^a \int_0^a xy \, dx \, dy \, dz = -\frac{1}{4} Ma^2 \end{aligned}$$

$$\begin{aligned} I_{11} &= \int_V \frac{M}{a^3} (x^2 + y^2 + z^2 - x^2) dV \\ &= \frac{M}{a^3} \int_0^a \int_0^a \int_0^a (y^2 + z^2) \, dx \, dy \, dz = \frac{2}{3} Ma^2 \end{aligned}$$

with the other components following by symmetry. We hence obtain

$$(I_{ij}) = \begin{pmatrix} \frac{2}{3} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{2}{3} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & \frac{2}{3} \end{pmatrix} Ma^2$$

A tedious calculation shows that the principal values are  $\frac{1}{6}Ma^2$ ,  $\frac{11}{12}Ma^2$ ,  $\frac{11}{12}Ma^2$ . The first principal axis is  $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$ , and the others are any two eigenvectors orthogonal to both of these – for example,

$$\begin{pmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix}, \quad \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}, \quad \begin{pmatrix} \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \\ -\frac{2}{\sqrt{6}} \end{pmatrix}$$

and we should check that this is orthonormal and right-handed.

In the basis given by these axes, the inertia tensor is just

$$\begin{pmatrix} \frac{1}{6} & 0 & 0 \\ 0 & \frac{11}{12} & 0 \\ 0 & 0 & \frac{11}{12} \end{pmatrix} Ma^2$$

and we note that the principal axes coincide with some symmetries of the physical object – here, for example, the first axis is parallel to the line between the origin and  $(a, a, a)$ .

## §8.8 Isotropic tensors

**Definition (Isotropic tensors).** An *isotropic tensor* is one whose *components* are the same in any basis.

We have already seen that  $\delta_{ij}$  and  $\varepsilon_{ijk}$  are isotropic. In fact, we can find all of them!

0. Any tensor of rank 0 is isotropic.

1. The only isotropic tensor of rank 1 is the zero vector.
2. The most general isotropic tensor of rank 2 is  $\lambda\delta_{ij}$  where  $\lambda$  is a scalar.
3. The most general isotropic tensor of rank 3 is  $\lambda\varepsilon_{ijk}$  where  $\lambda$  is a scalar.
4. The most general isotropic tensor of rank 4 is

$$\lambda\delta_{ij}\delta_{kl} + \mu\delta_{ik}\delta_{jl} + \nu\delta_{il}\delta_{jk}$$

where  $\lambda$ ,  $\mu$  and  $\nu$  are scalars.

The fourth-rank case need not be learned by heart – the second and third rank cases should, and are very important.

*Proof (rank 2).* Let  $T_{ij}$  be an isotropic second-rank tensor, and use the matrix version of the tensor transformation law,  $T' = RTR^\top$ . Let

$$R = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

that is, a rotation of  $\frac{\pi}{2}$  about the  $z$ -axis (note this is a rotation of  $-\frac{\pi}{2}$  in normal terms, but corresponds to rotating the *basis* by  $+\frac{\pi}{2}$ ).

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Then

$$\begin{pmatrix} T'_{11} & T'_{12} & T'_{13} \\ T'_{21} & T'_{22} & T'_{23} \\ T'_{31} & T'_{32} & T'_{33} \end{pmatrix} = R \begin{pmatrix} T_{11} & T_{12} & T_{13} \\ T_{21} & T_{22} & T_{23} \\ T_{31} & T_{32} & T_{33} \end{pmatrix} R^\top = \begin{pmatrix} T_{22} & -T_{21} & T_{23} \\ -T_{12} & T_{11} & -T_{13} \\ T_{32} & -T_{31} & T_{33} \end{pmatrix}$$

Since  $T'_{ij} = T_{ij}$ , we obtain

$$\begin{aligned} T_{11} &= T_{22} \\ T_{13} = T_{23} &= -T_{13} \implies T_{13} = T_{23} = 0 \\ T_{31} = T_{32} &= -T_{31} \implies T_{31} = T_{32} = 0 \end{aligned}$$

Similarly, a rotation of  $\frac{\pi}{2}$  about the  $y$ -axis gives

$$\begin{aligned} T_{11} &= T_{33} \\ T_{12} &= T_{32} = 0 \\ T_{21} &= T_{23} = 0 \end{aligned}$$

Hence, all off-diagonal entries are zero, and  $T_{11} = T_{22} = T_{33}$ . The result follows. ■

**Remark.** It is perhaps interesting that considering only two changes of basis suffices to show the result. Often, only a small number of symmetries are needed to make large simplifications.

The proof for third-rank tensors is just more long and painful.

What is the physical significance of isotropy? It essentially means that properties should be the same in all directions, and hence corresponds naturally to isotropic media.

## §8.9 Isotropic integrals

### Example

Let us attempt to calculate

$$I_{ij} = \int_{r < a} r x_i x_j dV$$

We can see that  $I_{ij}$  is a tensor, since it is a product of tensors followed by an integration. Furthermore, no ‘preferred direction’ is indicated in the integral, so the tensor must be isotropic.

Therefore, it must be the case that  $I_{ij} = \lambda \delta_{ij}$ . Hence, consider

$$\begin{aligned} 3\lambda = I_{ii} &= \int_{r < a} r x_i x_i dV \\ &= \int_{r=0}^a \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} r^3 r^2 \sin \theta dr d\theta d\phi \\ &= \frac{2}{3} \pi a^6 \end{aligned}$$

Hence

$$I_{ij} = \frac{2}{9} \pi a^6 \delta_{ij}$$

### Example

What about

$$\mathbf{B} = \int_{r < a} (\mathbf{x} \cdot \mathbf{b}) |\mathbf{x}| \mathbf{x} dV$$

which is *not* isotropic, since it depends on the direction of  $\mathbf{b}$ . Nonetheless, observe

$$\begin{aligned} B_i &= b_j \int_{r < a} x_j r x_i dV \\ &= b_j I_{ij} = b_j \frac{2}{9} \pi a^6 \delta_{ij} \\ \implies \mathbf{B} &= \frac{2}{9} \pi a^6 \mathbf{b} \end{aligned}$$

## §8.10 Tensor calculus

When we were defining differential operators, we did so with respect to the standard basis. But, had we used a different basis, we would hope that our results should transform correctly.

**Proposition 8.10**

Let  $\bullet$  be a tensor expression. Then

$$\frac{\partial \bullet}{\partial x'_i} = \frac{\partial \bullet}{\partial x_j} \frac{\partial x_j}{\partial x'_i} = R_{ij} \frac{\partial \bullet}{\partial x_j}$$

*Proof.* The first identity is just the chain rule – it remains only to establish that  $\frac{\partial x_j}{\partial x'_i} = R_{ij}$ . Observe

$$\begin{aligned} x_i &= R_{ji} x'_j \\ \Rightarrow \frac{\partial x_i}{\partial x'_k} &= R_{ji} \frac{\partial x'_j}{\partial x'_k} = R_{ji} \delta_{jk} = R_{ki} \end{aligned}$$

as required. ■

If we write  $\partial_i$  for the operator  $\frac{\partial}{\partial x_i}$ , then we obtain

$$\partial'_i = R_{ij} \partial_j$$

in other words,  $\nabla$  obeys the vector transformation law, and so is a vector differential operator.

Unsurprisingly, this allows us to show that all the vector differential operators transform in the right way. For example,

$$\begin{aligned} (\nabla \cdot \mathbf{F})' &= \nabla' \cdot \mathbf{F}' = \frac{\partial F'_i}{\partial x'_j} \\ &= R_{ij} \frac{\partial}{\partial x_i} (R_{ip} F_p) = R_{ij} R_{ip} \frac{\partial F_p}{\partial x_j} \\ &= \delta_{jp} \frac{\partial F_p}{\partial x_j} \\ &= \frac{\partial F_j}{\partial x_j} = \nabla \cdot \mathbf{F} \end{aligned}$$

The same results hold for  $\nabla f$  and  $\nabla \times \mathbf{F}$ .

A tensor field is a tensor which depends on position. If  $T_{pqr\dots}$  is a tensor field of rank  $m$ , then

$$\frac{\partial^n}{\partial x_i \partial x_j \partial x_k \dots} T_{pqr\dots}$$

is also a tensor field, of rank  $m + n$ .

**§8.10.1 The tensor divergence theorem (non-examinable)****Proposition**

For a tensor expression  $\bullet$  in a volume  $V$ ,

$$\int_V \partial \bullet x_i dV = \int_{\partial V} \bullet n_i dS$$

*Proof (rank 2).* Let  $F_i = a_j T_{ij}$ , where  $\mathbf{a}$  is an arbitrary constant vector. Then  $\mathbf{F}$  is a vector field, so we may apply the usual divergence theorem to obtain

$$\begin{aligned} \int_V \partial F_i x_i \, dV &= \int_{\partial V} \mathbf{F} \cdot d\mathbf{S} = \int_{\partial V} F_i n_i \, dS \\ &\implies a_j \int_V \end{aligned}$$

■